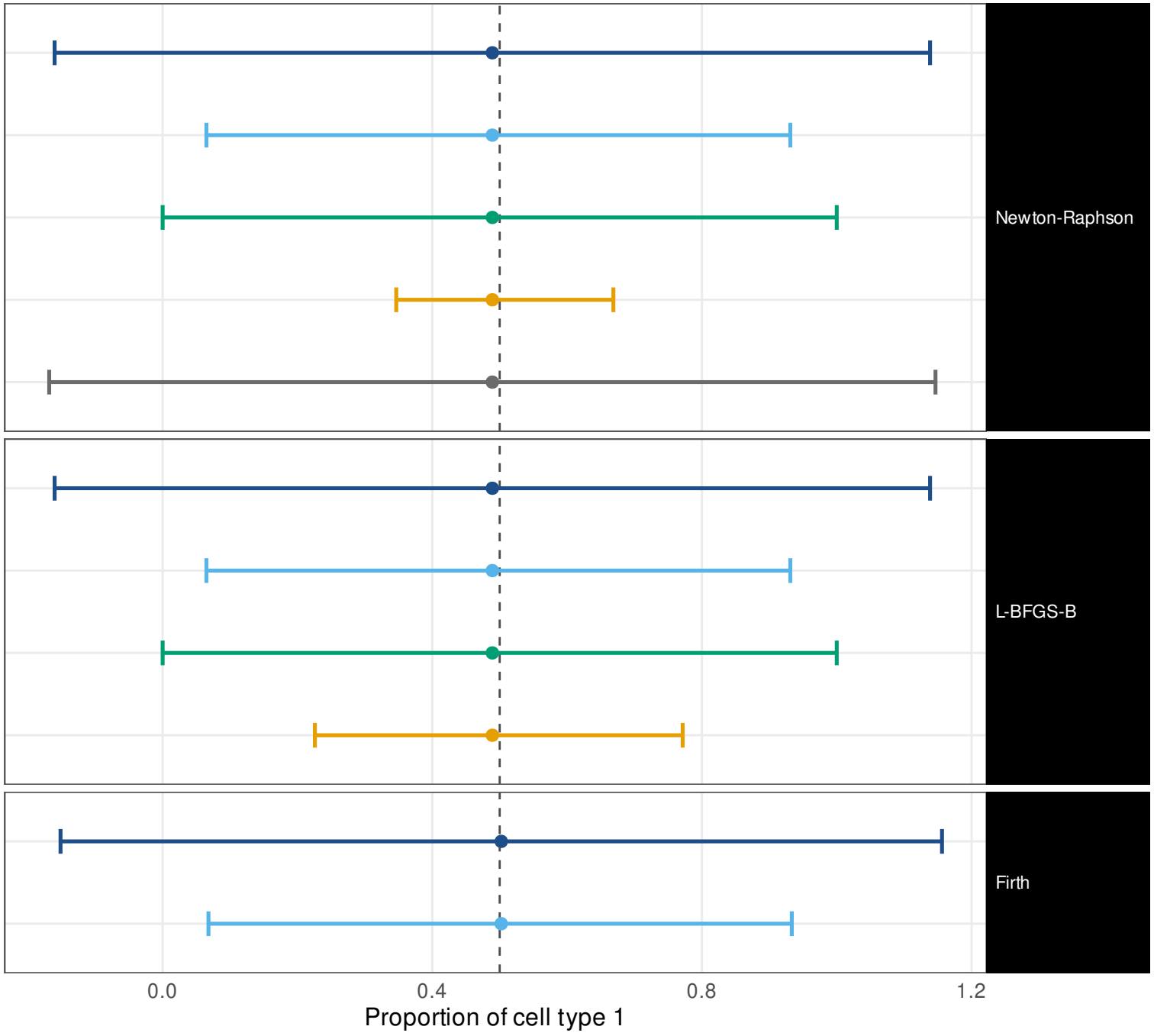
Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / homoscedastic / balanced / rho = (-0.8, -0.8)

```
Wald (delta)              1  [0.57,    1]   w =  1.3
ILR, mapped back          1  [0.57,    1]   w = 0.866
Profile likelihood        1  [0.57,    1]   w =    1
Parametric bootstrap    0.8  [0.38, 0.96]   w = 0.322
Oracle (Fisher at p*)     1  [0.57,    1]   w = 1.31

Wald (delta)              1  [0.57,    1]   w =  1.3
ILR, mapped back          1  [0.57,    1]   w = 0.866
Profile likelihood        1  [0.57,    1]   w =    1
Parametric bootstrap    0.8  [0.38, 0.96]   w = 0.546

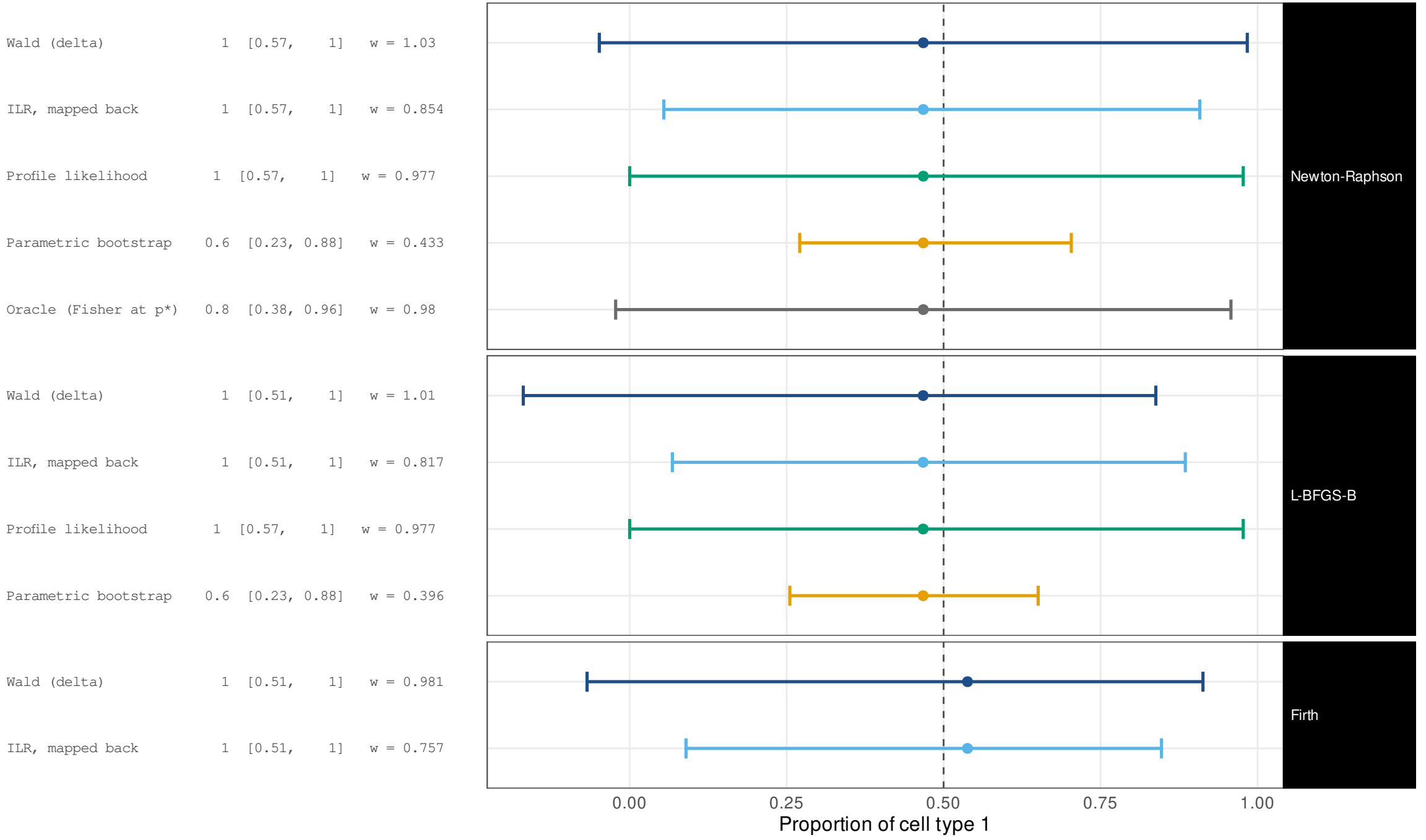
Wald (delta)              1  [0.57,    1]   w = 1.31
ILR, mapped back          1  [0.57,    1]   w = 0.865
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

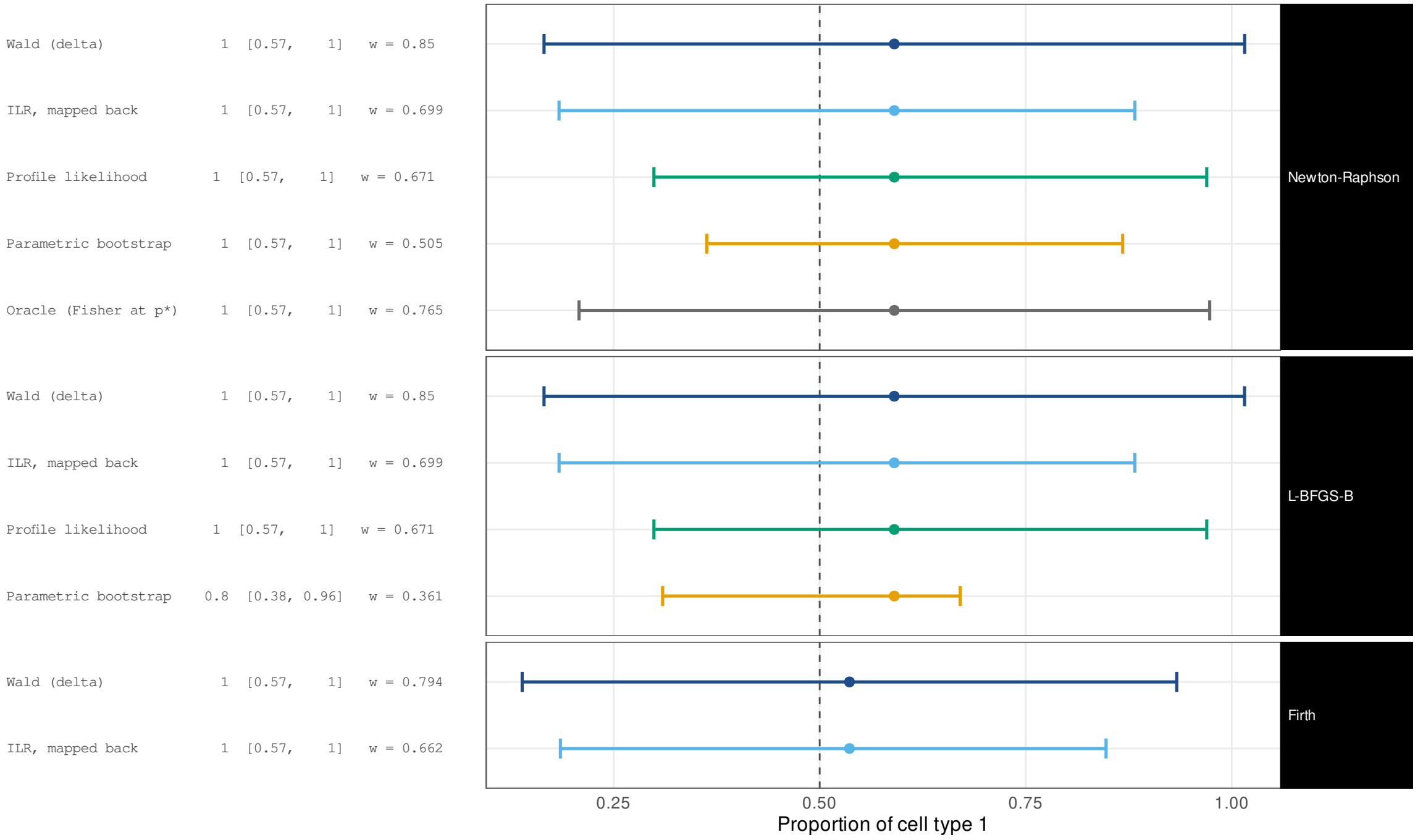
small centroid gap / homoscedastic / balanced / rho = (+0.0, +0.0)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / homoscedastic / balanced / rho = (-0.8, +0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

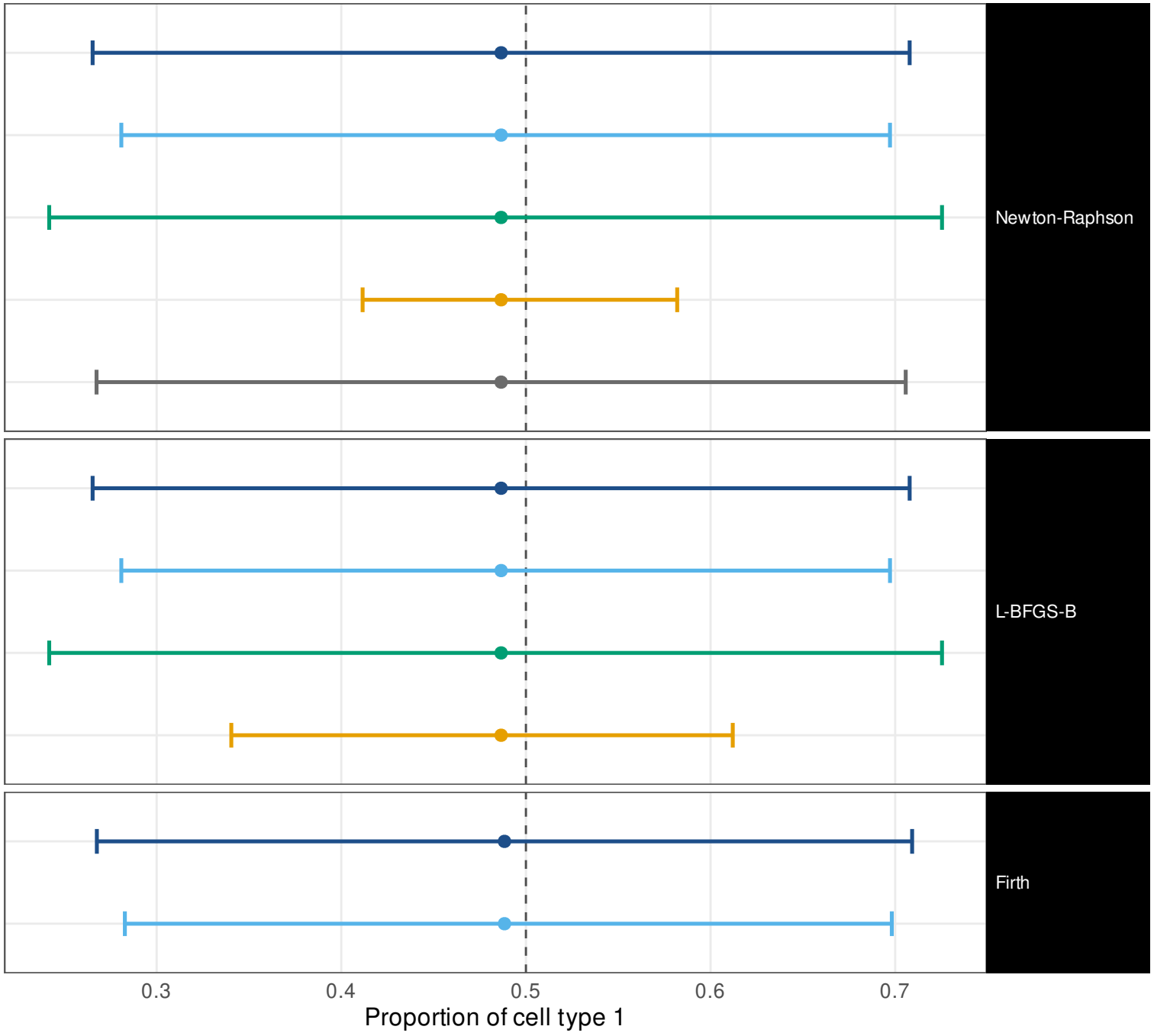
Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / homoscedastic / balanced / rho = (+0.8, +0.8)

```
Wald (delta)              1  [0.57,    1]   w = 0.443
ILR, mapped back          1  [0.57,    1]   w = 0.416
Profile likelihood        1  [0.57,    1]   w = 0.484
Parametric bootstrap    0.8  [0.38, 0.96]   w = 0.17
Oracle (Fisher at p*)     1  [0.57,    1]   w = 0.438

Wald (delta)              1  [0.57,    1]   w = 0.443
ILR, mapped back          1  [0.57,    1]   w = 0.416
Profile likelihood        1  [0.57,    1]   w = 0.484
Parametric bootstrap    0.8  [0.38, 0.96]   w = 0.272

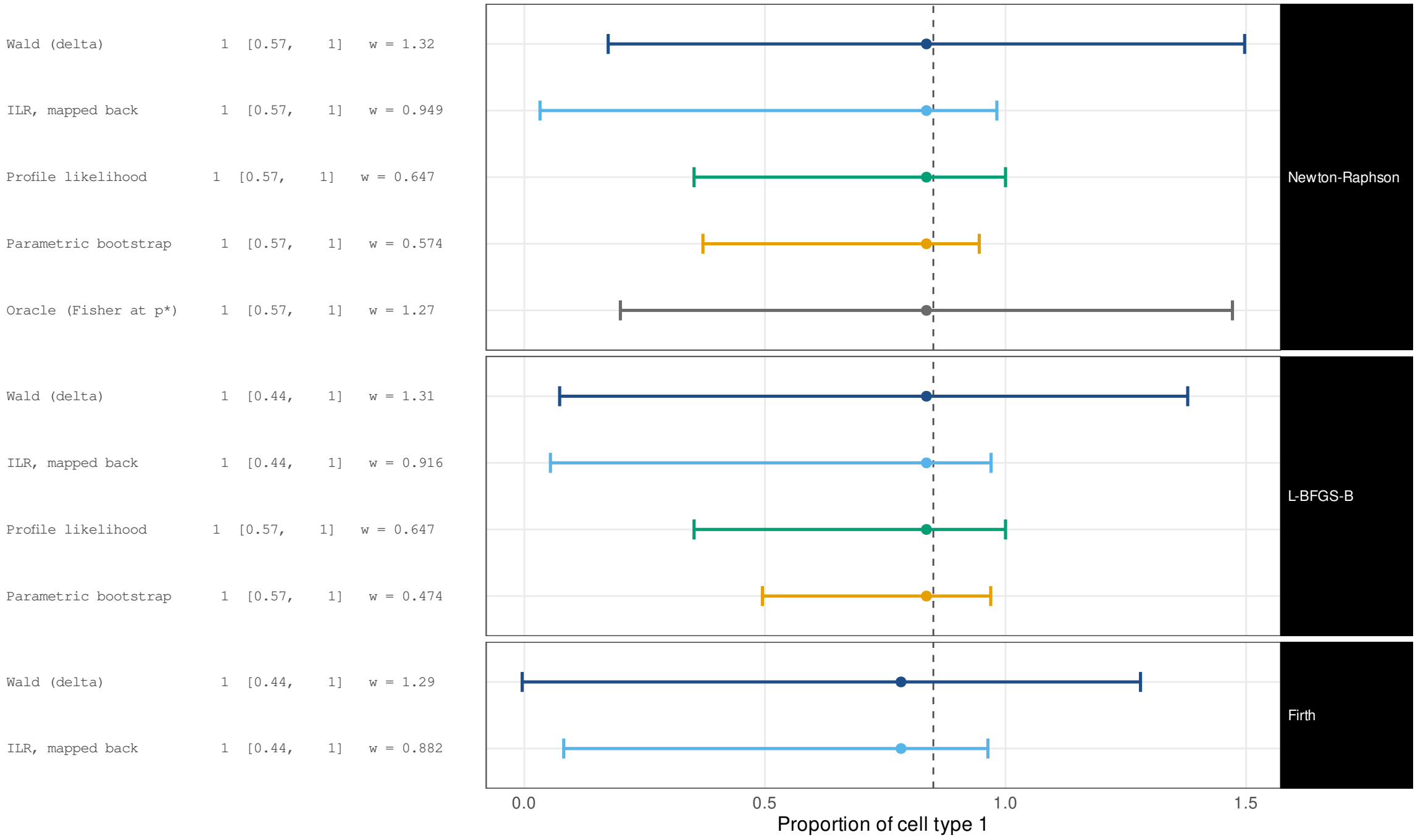
Wald (delta)              1  [0.57,    1]   w = 0.442
ILR, mapped back          1  [0.57,    1]   w = 0.415
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / homoscedastic / moderately unbalanced / rho = (-0.8, -0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

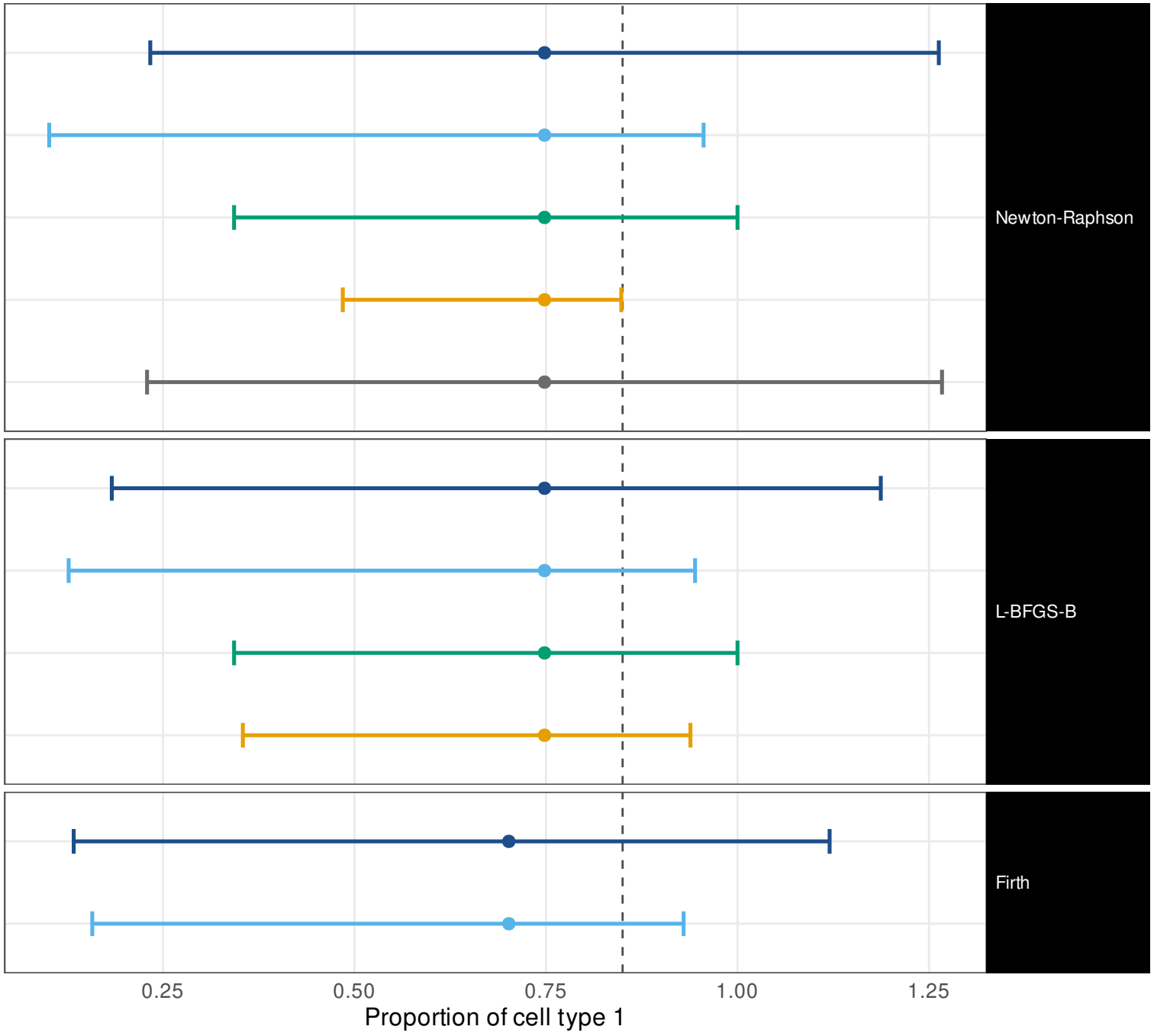
Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / homoscedastic / moderately unbalanced / rho = (+0.0, +0.0)

```
Wald (delta)              1  [0.57,    1]   w = 1.03
ILR, mapped back          1  [0.57,    1]   w = 0.854
Profile likelihood        1  [0.57,    1]   w = 0.657
Parametric bootstrap    0.6  [0.23, 0.88]   w = 0.364
Oracle (Fisher at p*)     1  [0.57,    1]   w = 1.04

Wald (delta)              1  [0.51,    1]   w =    1
ILR, mapped back          1  [0.51,    1]   w = 0.818
Profile likelihood        1  [0.57,    1]   w = 0.657
Parametric bootstrap    0.8  [0.38, 0.96]   w = 0.584

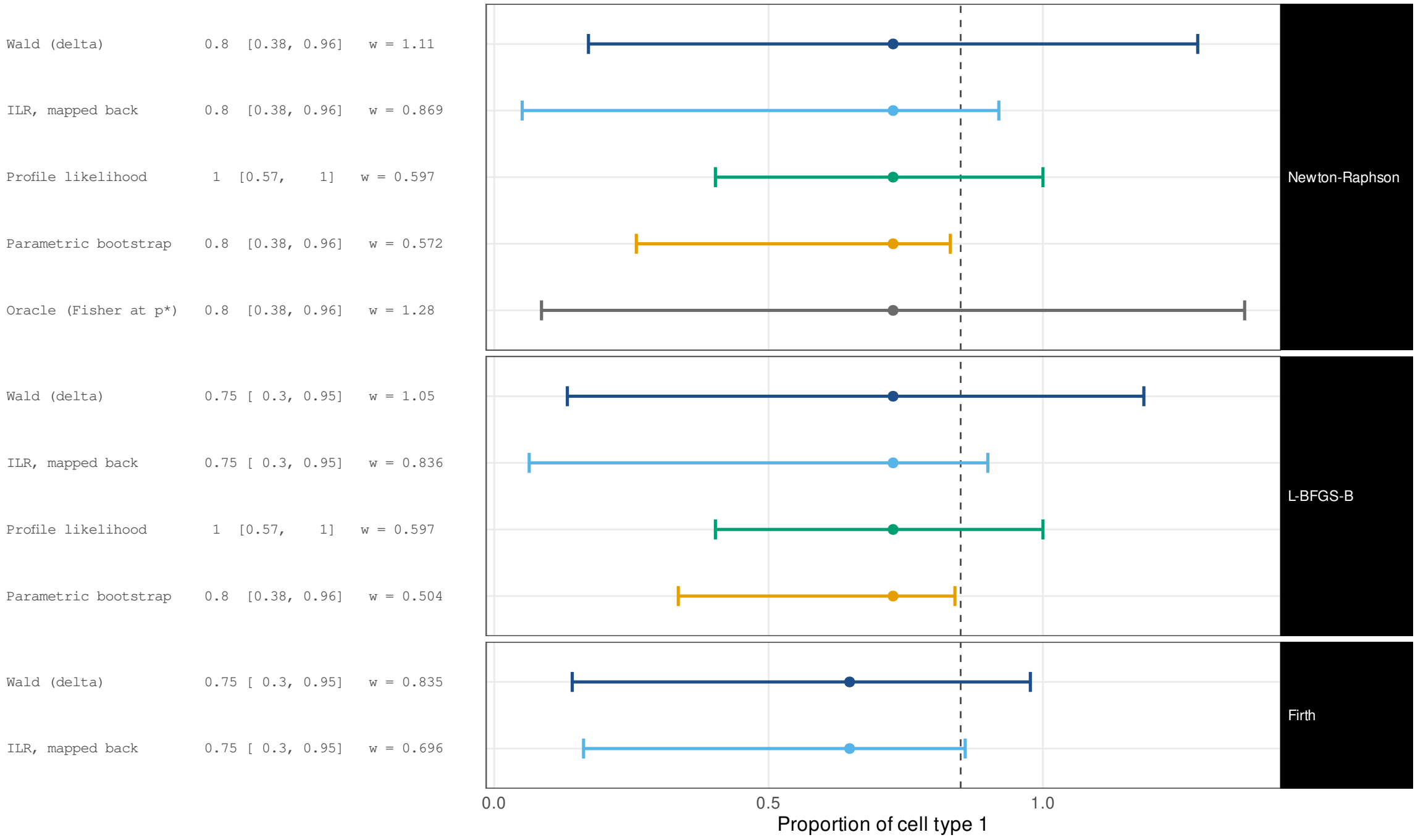
Wald (delta)              1  [0.51,    1]   w = 0.987
ILR, mapped back          1  [0.51,    1]   w = 0.772
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

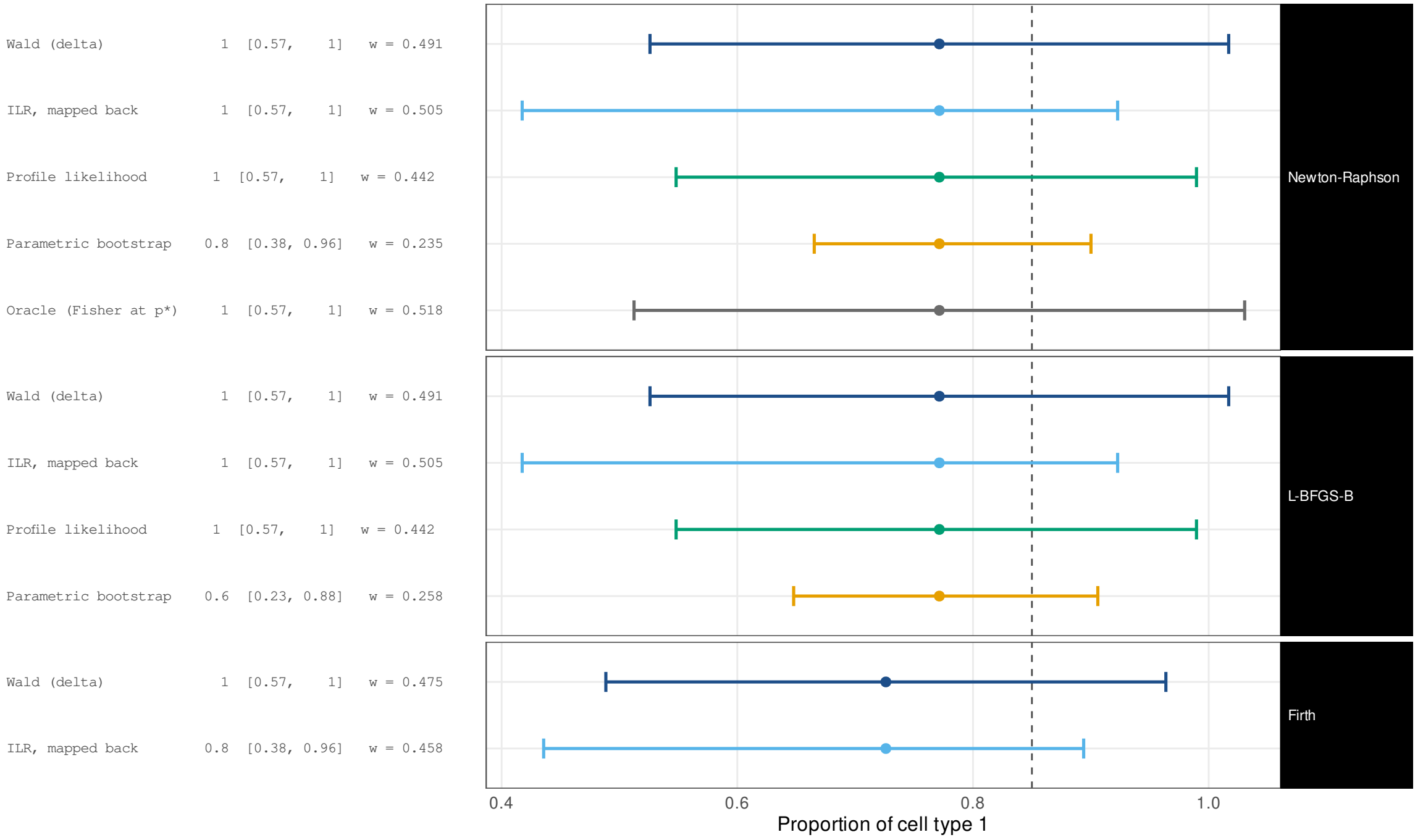
small centroid gap / homoscedastic / moderately unbalanced / rho = (-0.8, +0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / homoscedastic / moderately unbalanced / rho = (+0.8, +0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

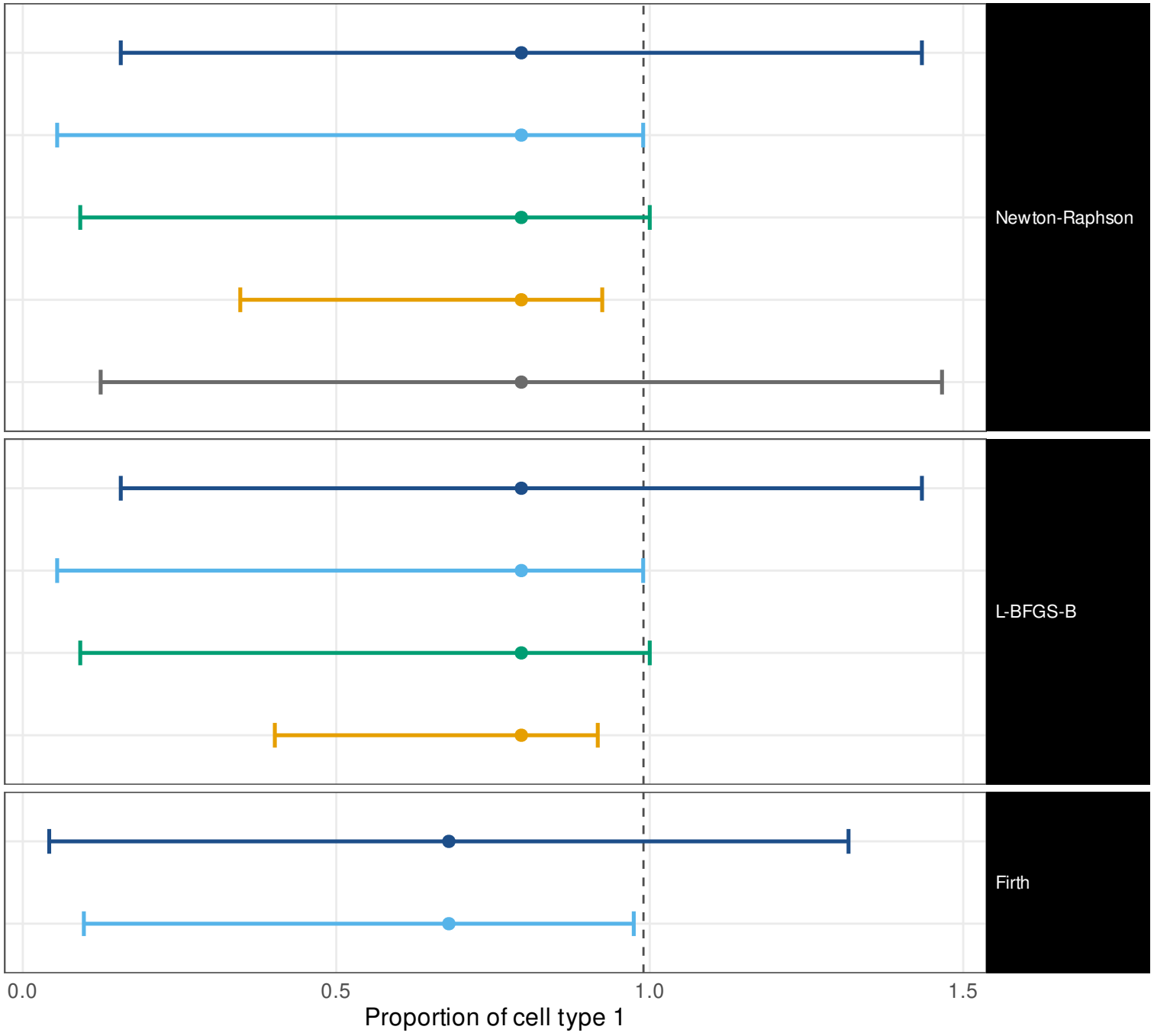
Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / homoscedastic / highly unbalanced / rho = (-0.8, -0.8)

```
Wald (delta)              1  [0.57,    1]   w = 1.28
ILR, mapped back        0.6  [0.23, 0.88]   w = 0.935
Profile likelihood        1  [0.57,    1]   w = 0.908
Parametric bootstrap      0  [   0, 0.43]   w = 0.577
Oracle (Fisher at p*)     1  [0.57,    1]   w = 1.34

Wald (delta)              1  [0.57,    1]   w = 1.28
ILR, mapped back        0.6  [0.23, 0.88]   w = 0.935
Profile likelihood        1  [0.57,    1]   w = 0.908
Parametric bootstrap    0.6  [0.23, 0.88]   w = 0.515

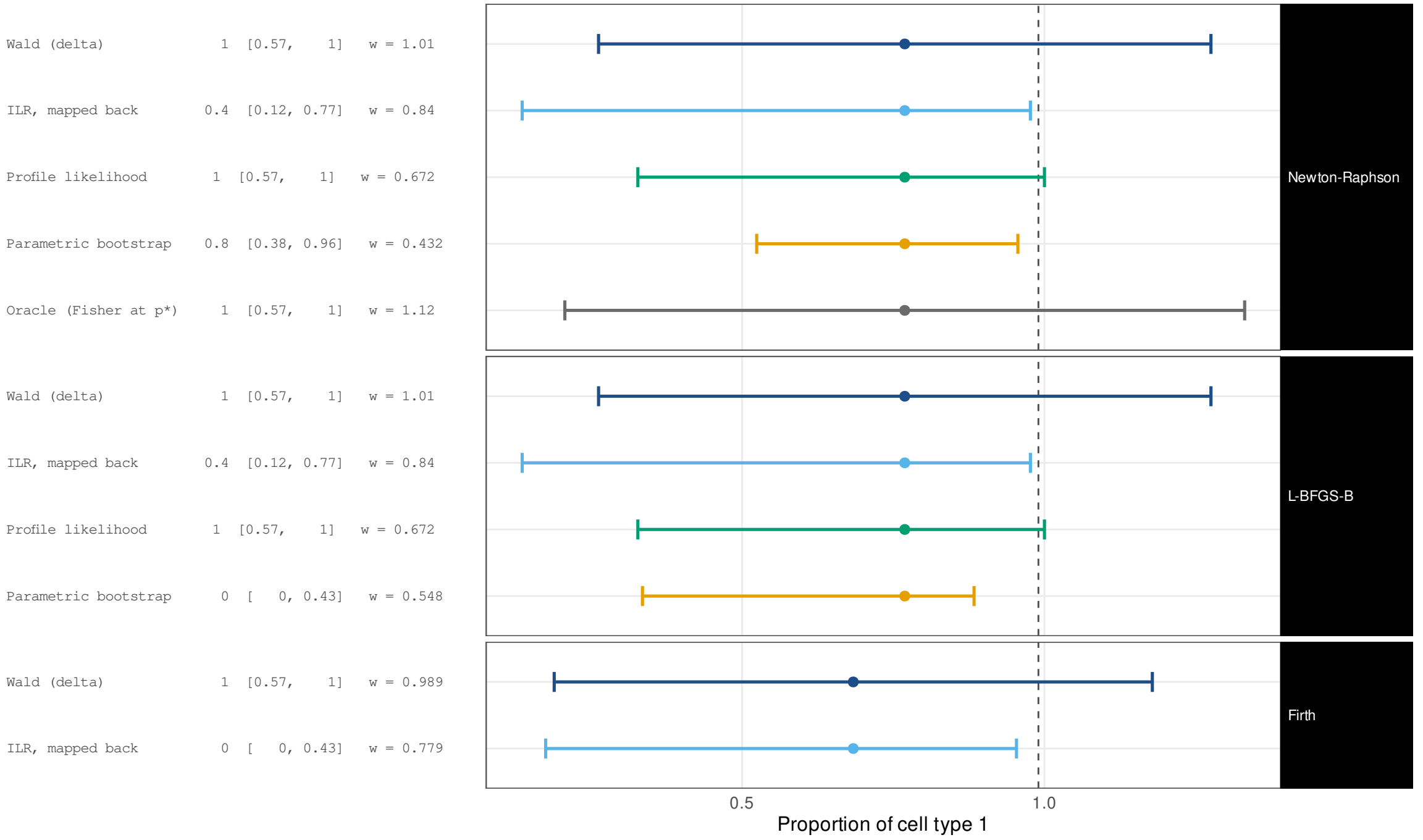
Wald (delta)              1  [0.57,    1]   w = 1.27
ILR, mapped back          0  [   0, 0.43]   w = 0.877
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

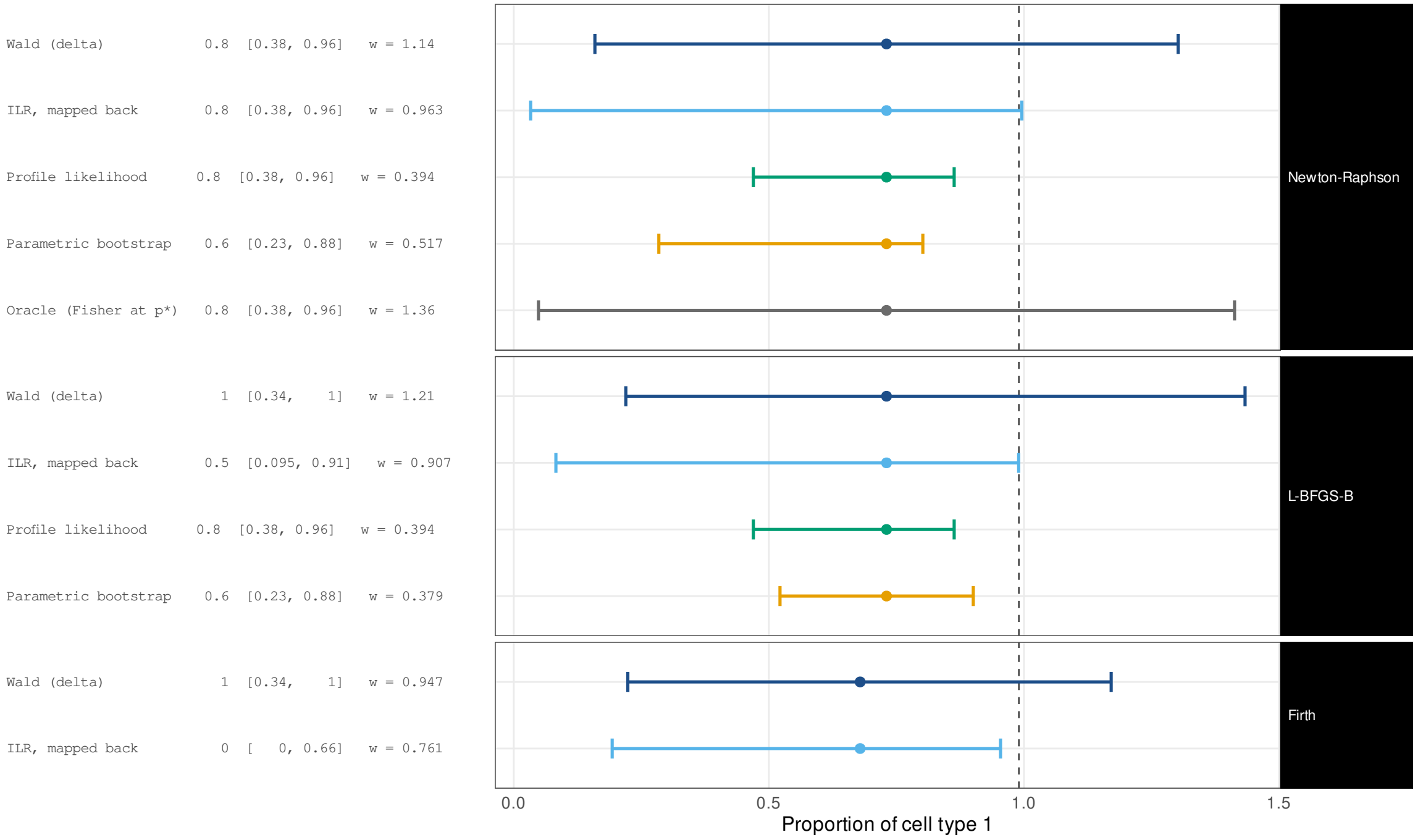
small centroid gap / homoscedastic / highly unbalanced / rho = (+0.0, +0.0)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / homoscedastic / highly unbalanced / rho = (-0.8, +0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

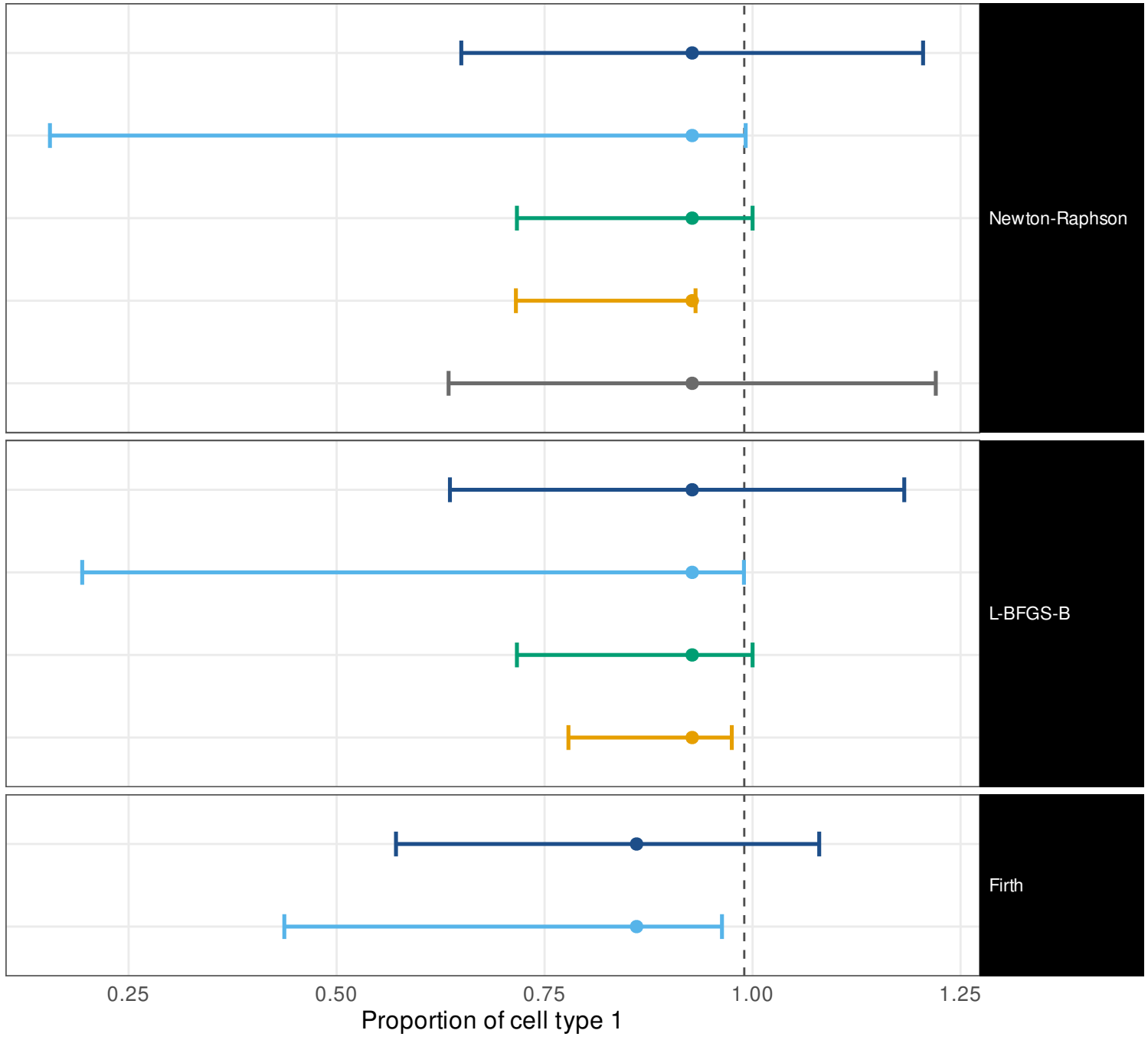
Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / homoscedastic / highly unbalanced / rho = (+0.8, +0.8)

```
Wald (delta)              1  [0.57,     1]   w = 0.555
ILR, mapped back        0.8  [0.38, 0.96]   w = 0.836
Profile likelihood        1  [0.57,     1]   w = 0.283
Parametric bootstrap    0.2  [0.036, 0.62]   w = 0.216
Oracle (Fisher at p*)     1  [0.57,     1]   w = 0.586

Wald (delta)              1  [0.51,     1]   w = 0.546
ILR, mapped back       0.75  [ 0.3, 0.95]   w = 0.795
Profile likelihood        1  [0.57,     1]   w = 0.283
Parametric bootstrap    0.8  [0.38, 0.96]   w = 0.196

Wald (delta)              1  [0.51,     1]   w = 0.509
ILR, mapped back          0  [    0, 0.49]   w = 0.526
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

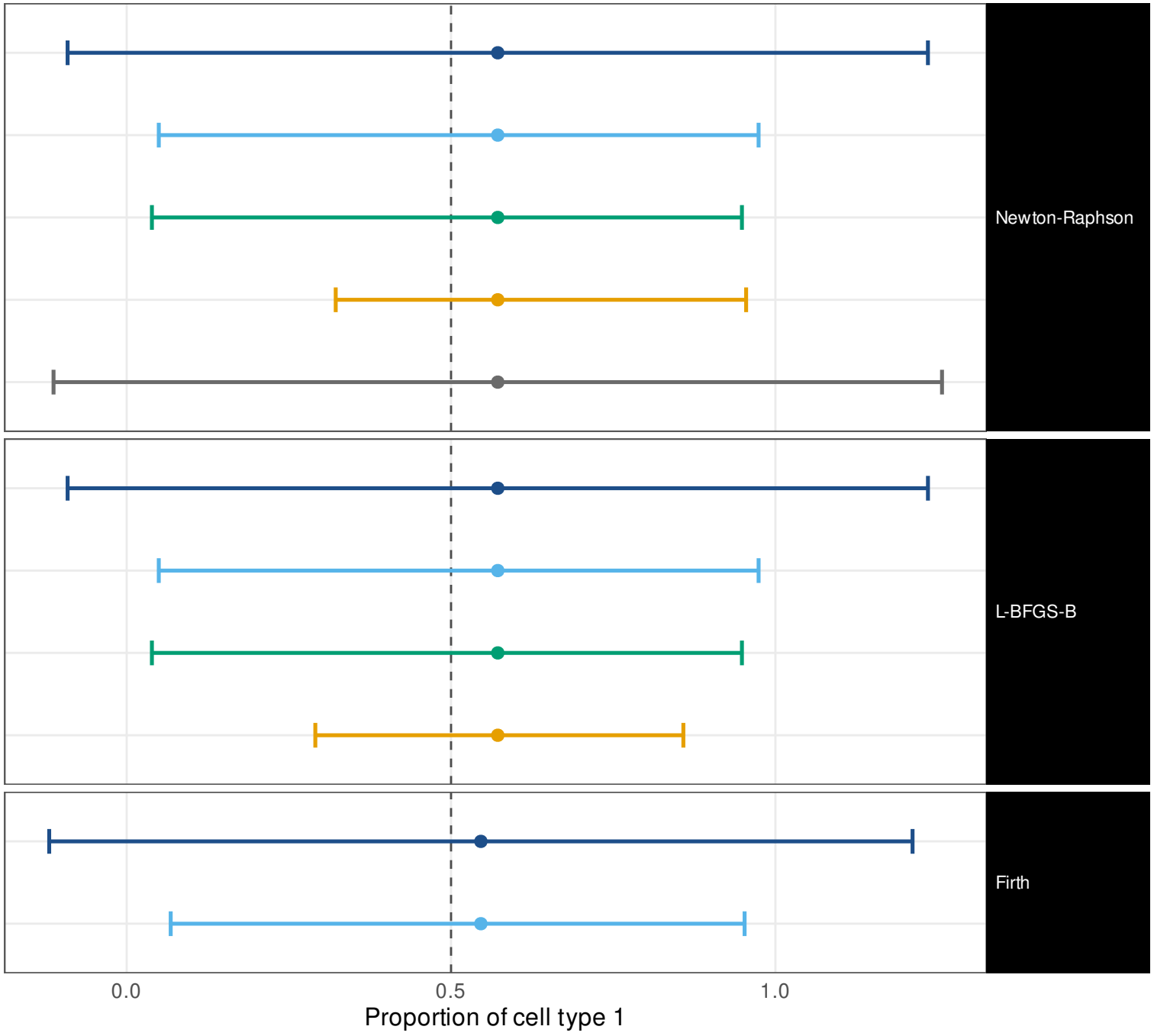
Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / heteroscedastic / balanced / rho = (-0.8, -0.8)

```
Wald (delta)              1  [0.57,    1]   w = 1.33
ILR, mapped back          1  [0.57,    1]   w = 0.925
Profile likelihood        1  [0.57,    1]   w = 0.91
Parametric bootstrap    0.4  [0.12, 0.77]   w = 0.633
Oracle (Fisher at p*)     1  [0.57,    1]   w = 1.37

Wald (delta)              1  [0.57,    1]   w = 1.33
ILR, mapped back          1  [0.57,    1]   w = 0.925
Profile likelihood        1  [0.57,    1]   w = 0.91
Parametric bootstrap      1  [0.57,    1]   w = 0.567

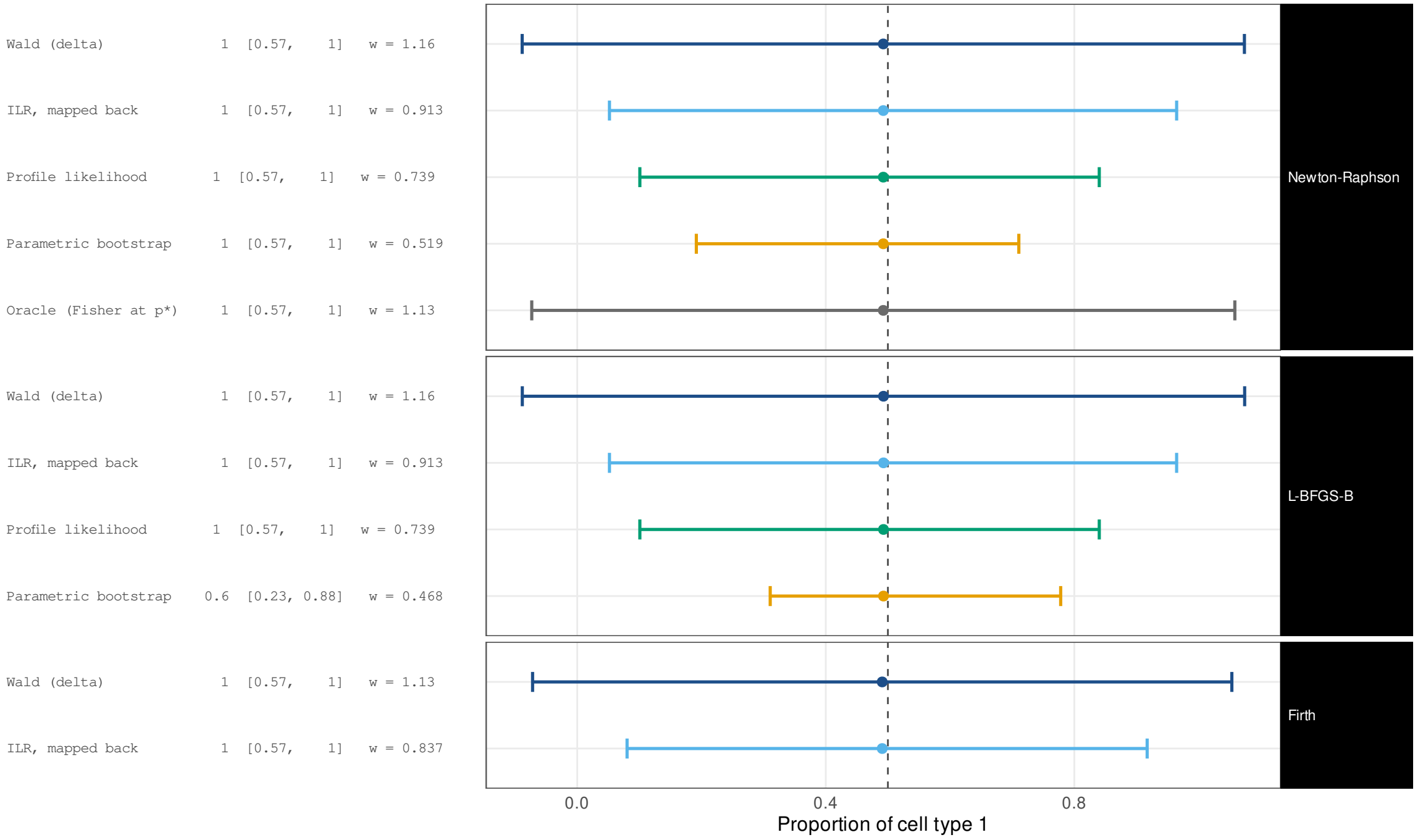
Wald (delta)              1  [0.57,    1]   w = 1.33
ILR, mapped back          1  [0.57,    1]   w = 0.885
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

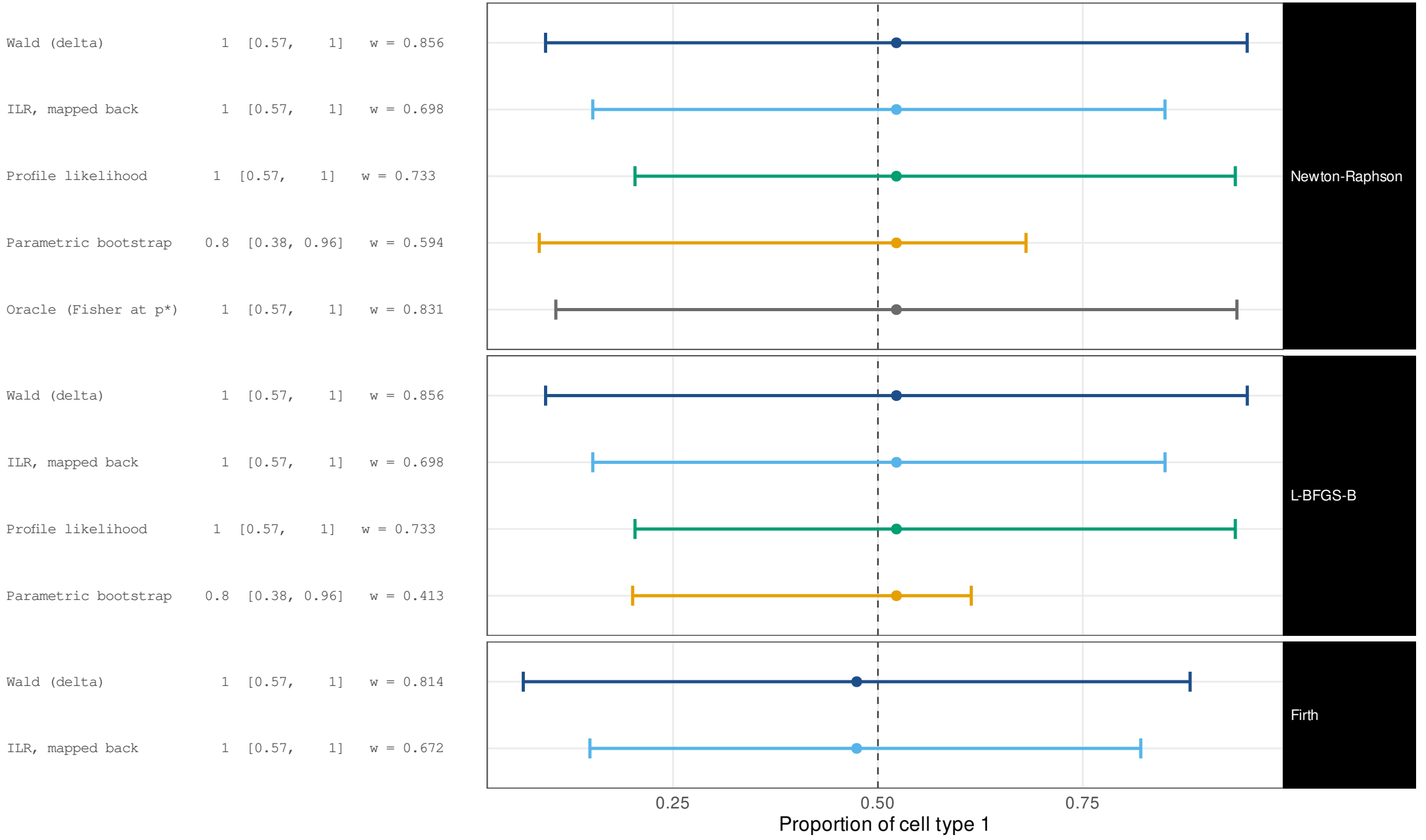
small centroid gap / heteroscedastic / balanced / rho = (+0.0, +0.0)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / heteroscedastic / balanced / rho = (-0.8, +0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

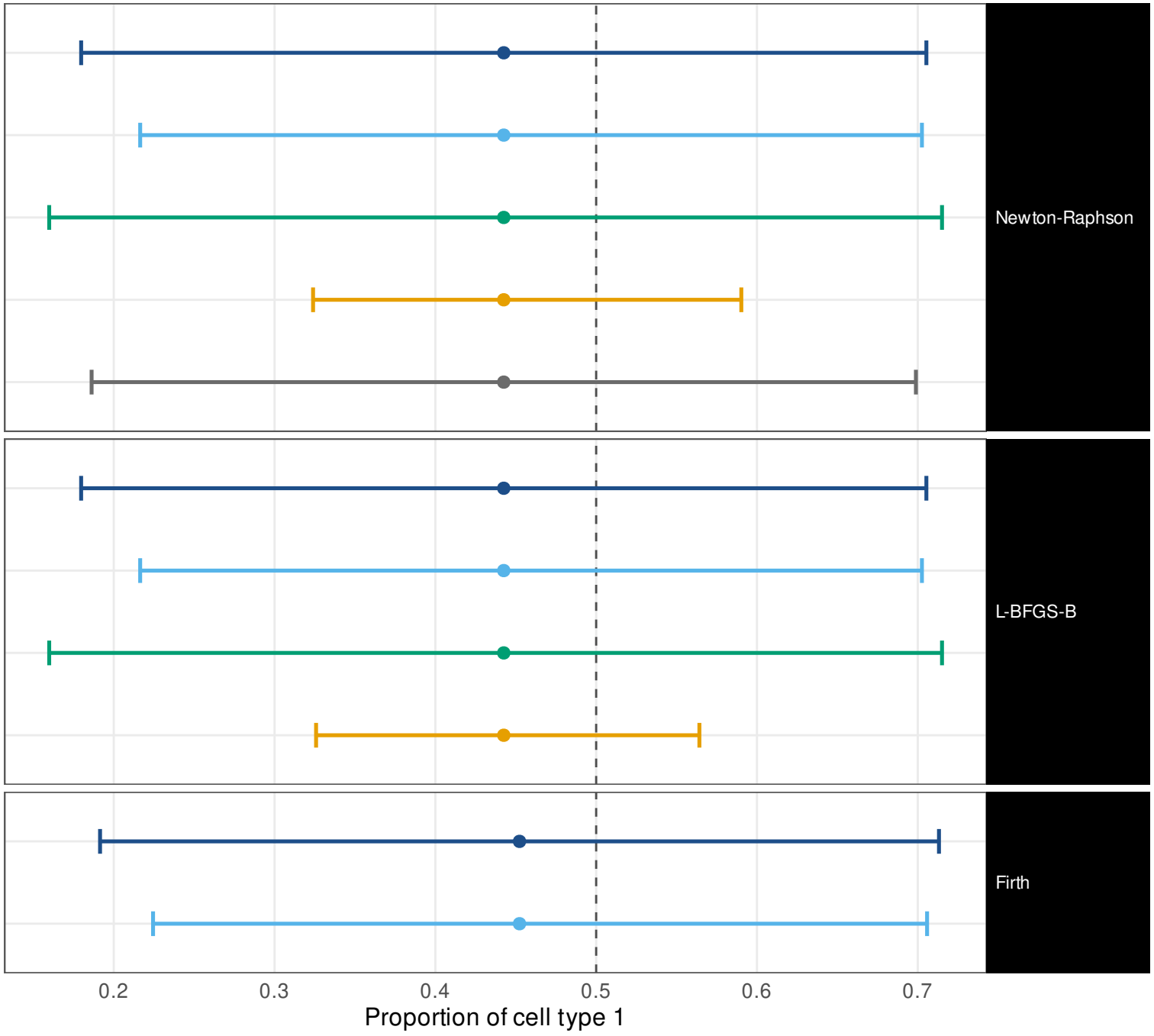
Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / heteroscedastic / balanced / rho = (+0.8, +0.8)

```
Wald (delta)              1  [0.57,    1]   w = 0.526
ILR, mapped back          1  [0.57,    1]   w = 0.486
Profile likelihood        1  [0.57,    1]   w = 0.555
Parametric bootstrap    0.8  [0.38, 0.96]   w = 0.266
Oracle (Fisher at p*)     1  [0.57,    1]   w = 0.513

Wald (delta)              1  [0.57,    1]   w = 0.526
ILR, mapped back          1  [0.57,    1]   w = 0.486
Profile likelihood        1  [0.57,    1]   w = 0.555
Parametric bootstrap    0.6  [0.23, 0.88]   w = 0.238

Wald (delta)              1  [0.57,    1]   w = 0.522
ILR, mapped back          1  [0.57,    1]   w = 0.481
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

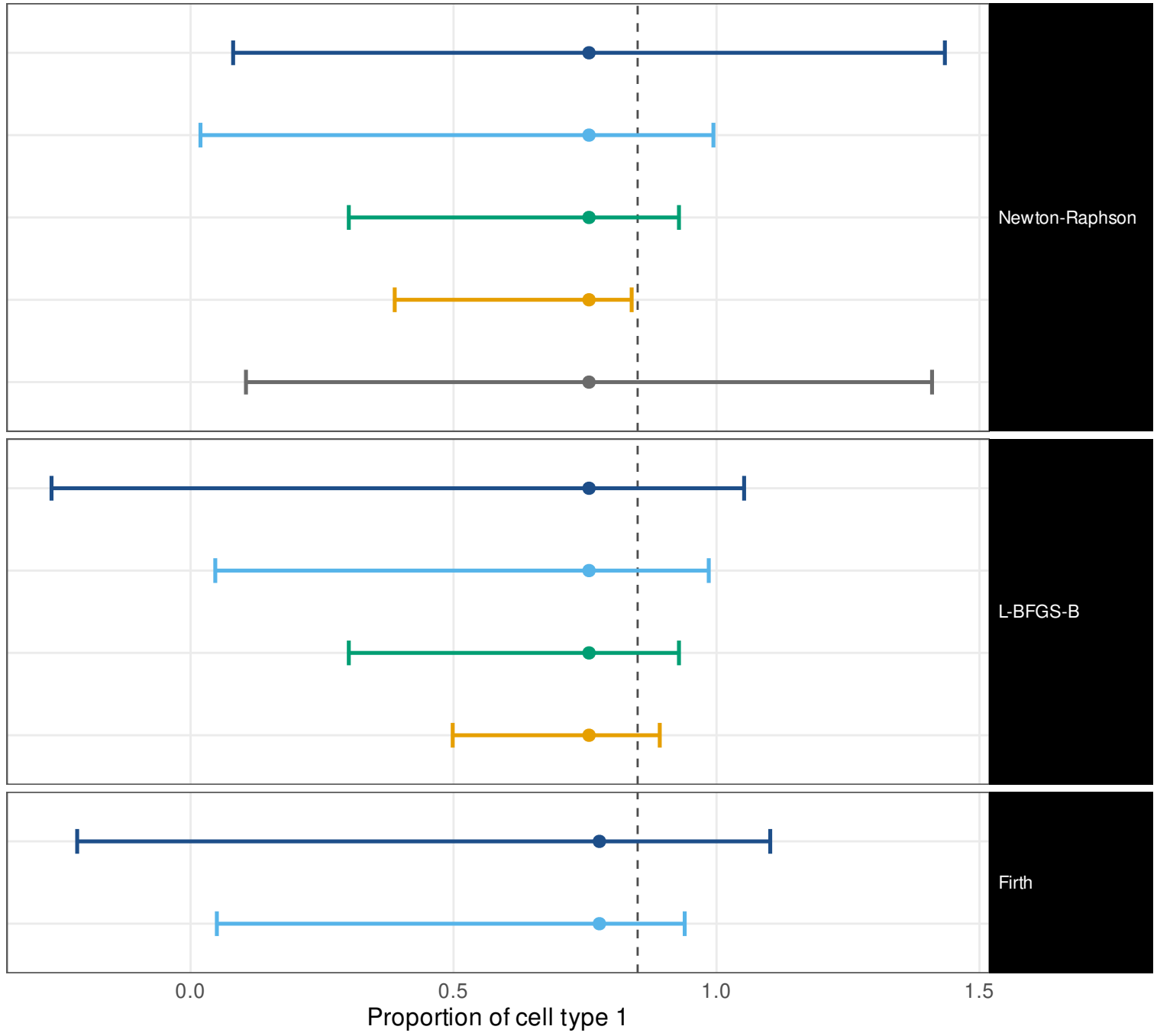
Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / heteroscedastic / moderately unbalanced / rho = (-0.8, -0.8)

```
Wald (delta)            0.8   [0.38, 0.96]   w = 1.35
ILR, mapped back          1   [0.57,    1]   w = 0.975
Profile likelihood      0.8   [0.38, 0.96]   w = 0.628
Parametric bootstrap    0.6   [0.23, 0.88]   w = 0.451
Oracle (Fisher at p*)   0.8   [0.38, 0.96]   w =  1.3

Wald (delta)            0.5   [0.095, 0.91]   w = 1.32
ILR, mapped back          1   [0.34,    1]   w = 0.938
Profile likelihood      0.8   [0.38, 0.96]   w = 0.628
Parametric bootstrap    0.6   [0.23, 0.88]   w = 0.394

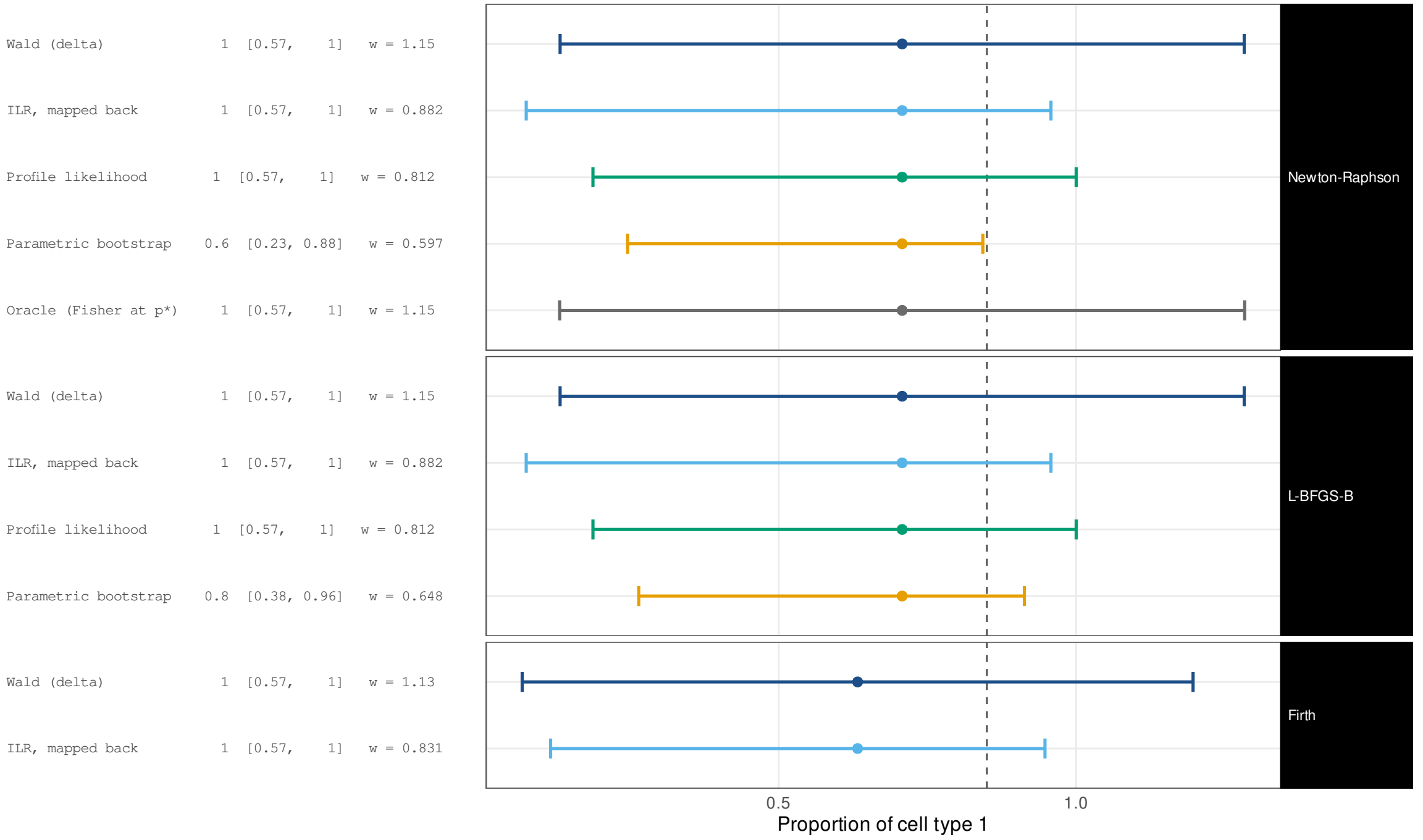
Wald (delta)              1   [0.34,    1]   w = 1.32
ILR, mapped back          1   [0.34,    1]   w = 0.89
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

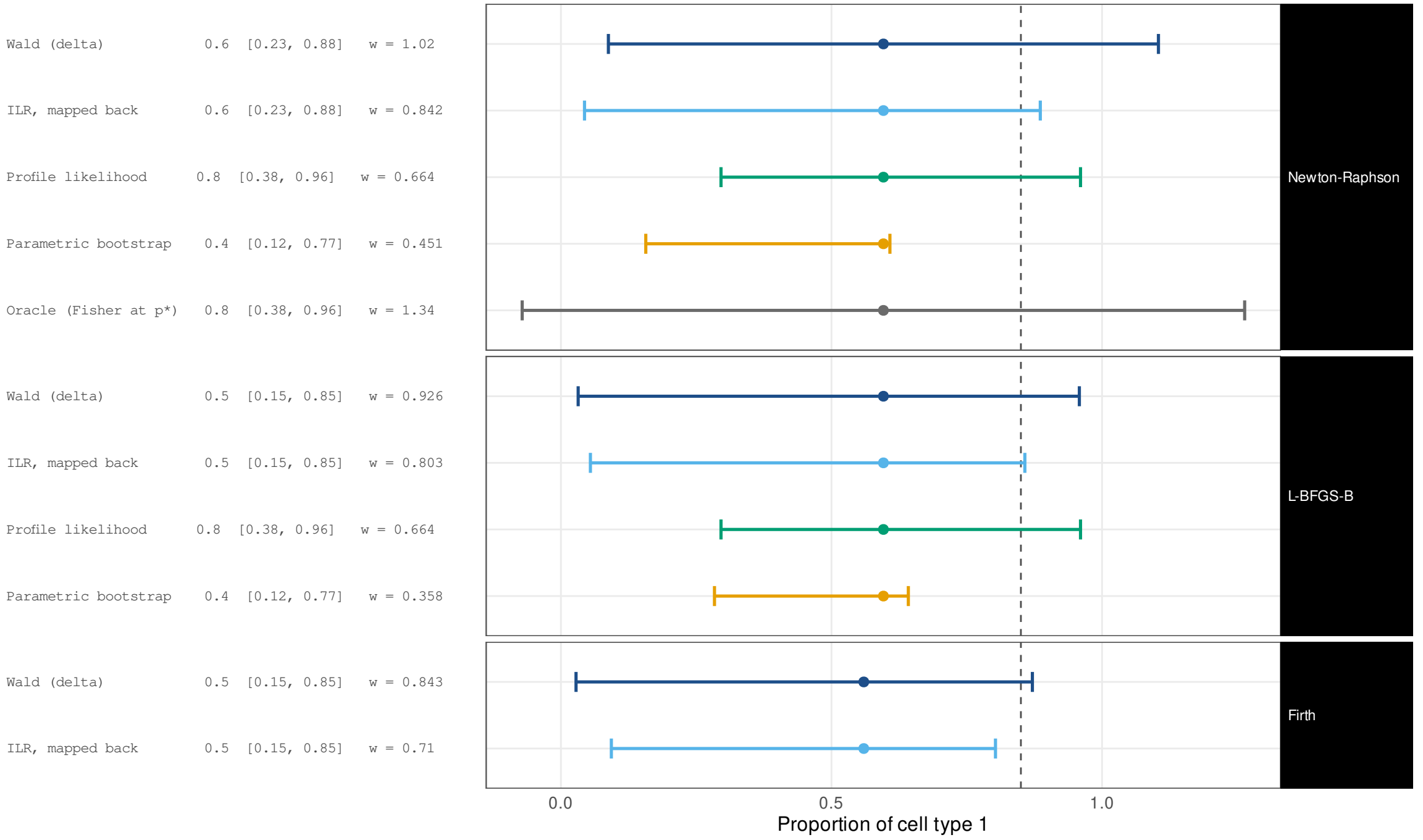
small centroid gap / heteroscedastic / moderately unbalanced / rho = (+0.0, +0.0)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / heteroscedastic / moderately unbalanced / rho = (-0.8, +0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

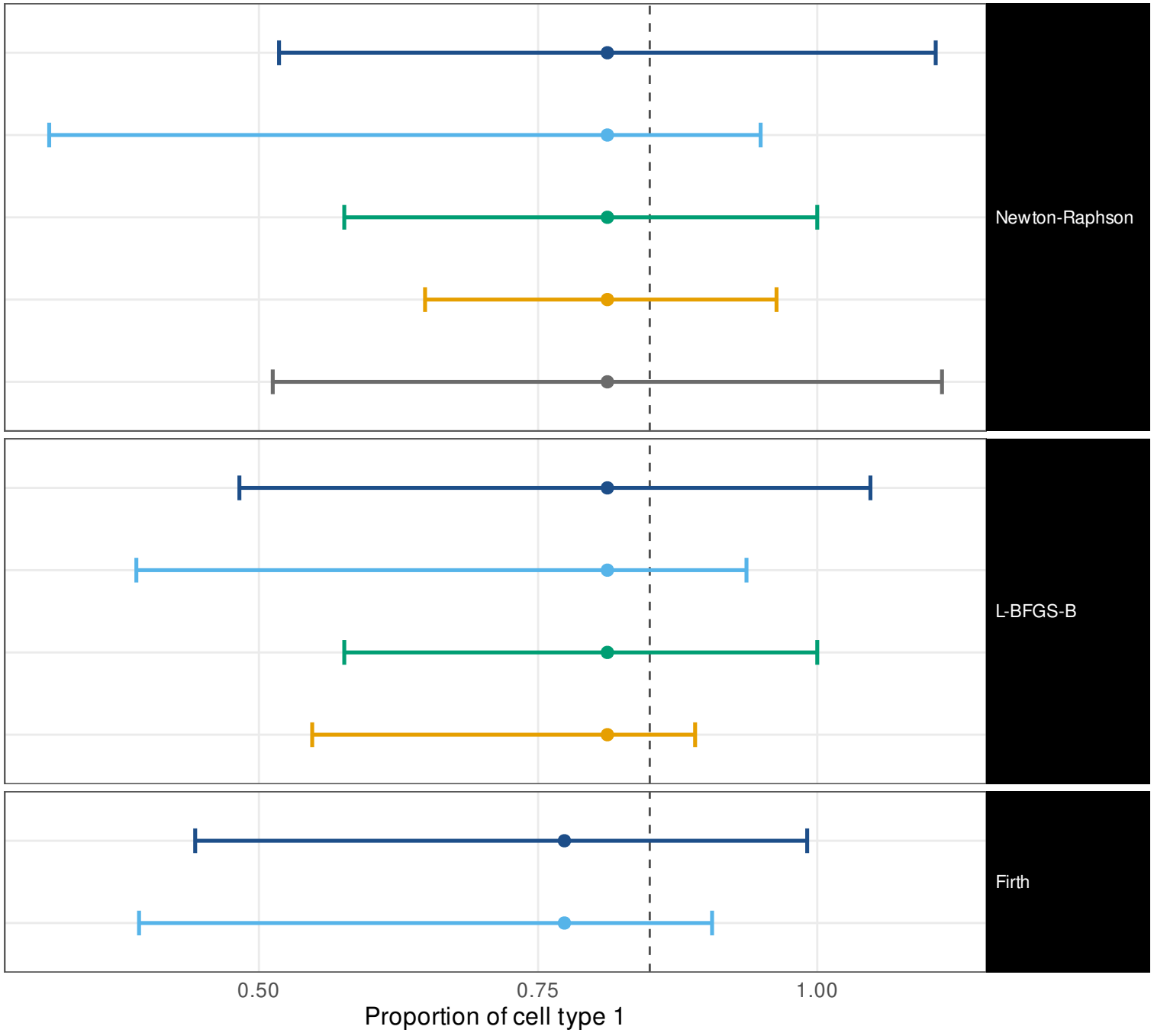
Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / heteroscedastic / moderately unbalanced / rho = (+0.8, +0.8)

```
Wald (delta)              1  [0.57,    1]   w = 0.588
ILR, mapped back          1  [0.57,    1]   w = 0.637
Profile likelihood        1  [0.57,    1]   w = 0.424
Parametric bootstrap    0.8  [0.38, 0.96]   w = 0.315
Oracle (Fisher at p*)     1  [0.57,    1]   w = 0.599

Wald (delta)              1  [0.51,    1]   w = 0.565
ILR, mapped back          1  [0.51,    1]   w = 0.546
Profile likelihood        1  [0.57,    1]   w = 0.424
Parametric bootstrap    0.8  [0.38, 0.96]   w = 0.343

Wald (delta)              1  [0.51,    1]   w = 0.548
ILR, mapped back          1  [0.51,    1]   w = 0.513
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

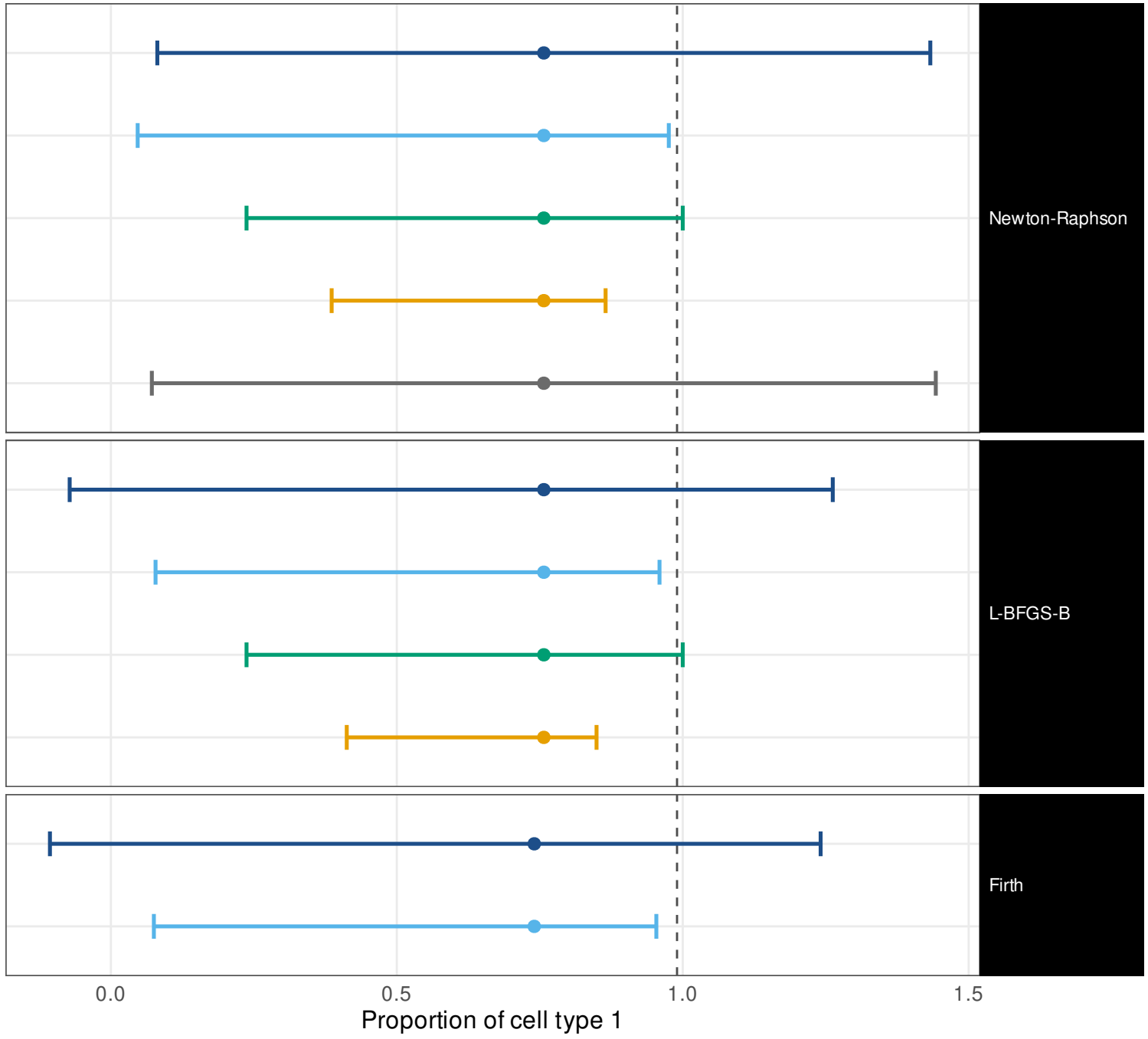
Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / heteroscedastic / highly unbalanced / rho = (-0.8, -0.8)

```
Wald (delta)              1  [0.57,     1]   w = 1.35
ILR, mapped back        0.4  [0.12, 0.77]   w = 0.929
Profile likelihood        1  [0.57,     1]   w = 0.763
Parametric bootstrap    0.2  [0.036, 0.62]   w = 0.479
Oracle (Fisher at p*)     1  [0.57,     1]   w = 1.37

Wald (delta)              1  [0.44,     1]   w = 1.33
ILR, mapped back          0  [    0, 0.56]   w = 0.881
Profile likelihood        1  [0.57,     1]   w = 0.763
Parametric bootstrap    0.2  [0.036, 0.62]   w = 0.437

Wald (delta)              1  [0.44,     1]   w = 1.35
ILR, mapped back          0  [    0, 0.56]   w = 0.879
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

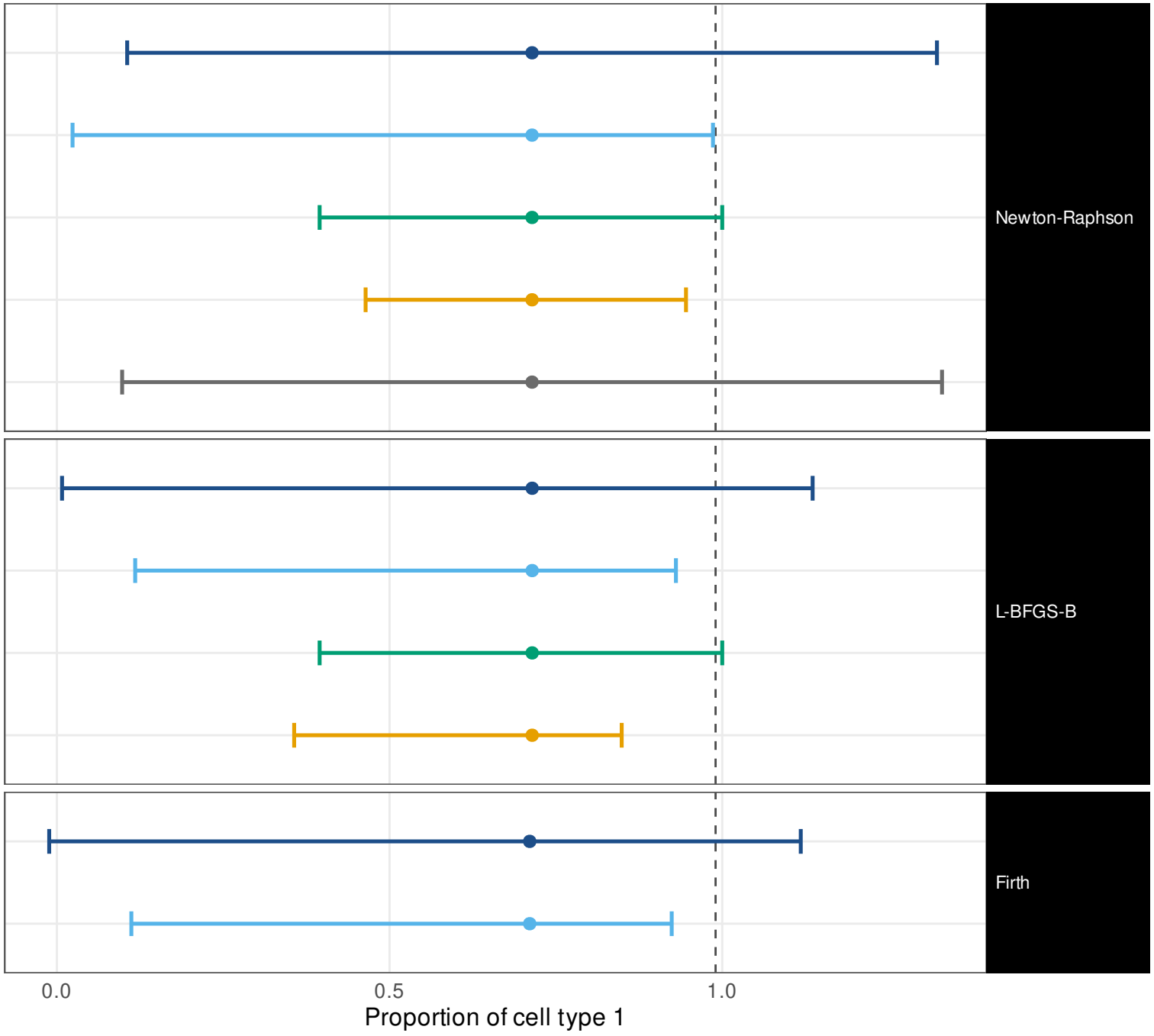
Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / heteroscedastic / highly unbalanced / rho = (+0.0, +0.0)

```
Wald (delta)            0.8  [0.38, 0.96]   w = 1.22
ILR, mapped back        0.8  [0.38, 0.96]   w = 0.963
Profile likelihood        1  [0.57,    1]   w = 0.605
Parametric bootstrap    0.4  [0.12, 0.77]   w = 0.482
Oracle (Fisher at p*)   0.8  [0.38, 0.96]   w = 1.23

Wald (delta)              1  [0.21,    1]   w = 1.13
ILR, mapped back          0  [   0, 0.79]   w = 0.813
Profile likelihood        1  [0.57,    1]   w = 0.605
Parametric bootstrap    0.4  [0.12, 0.77]   w = 0.492

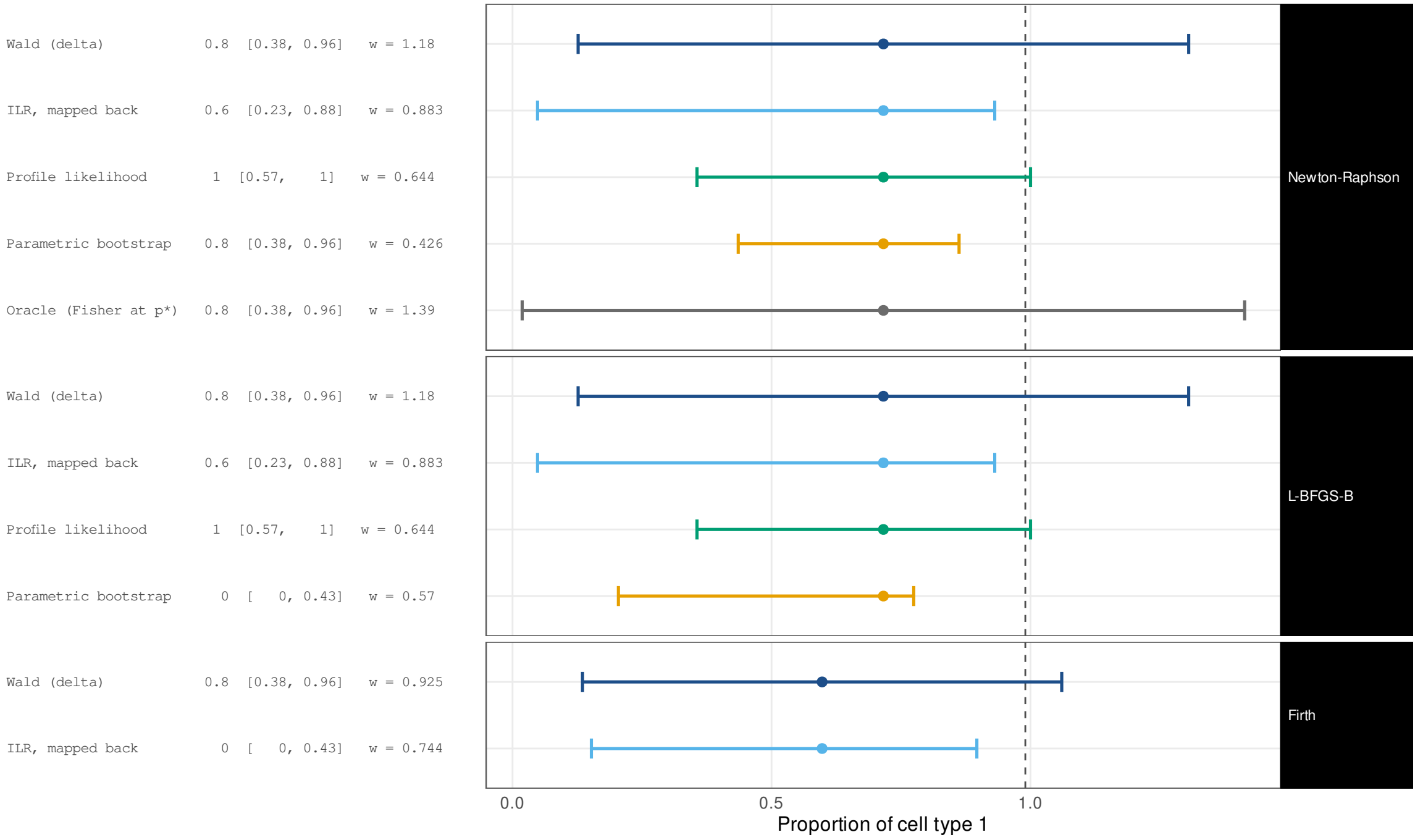
Wald (delta)              1  [0.21,    1]   w = 1.13
ILR, mapped back          0  [   0, 0.79]   w = 0.812
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / heteroscedastic / highly unbalanced / rho = (-0.8, +0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

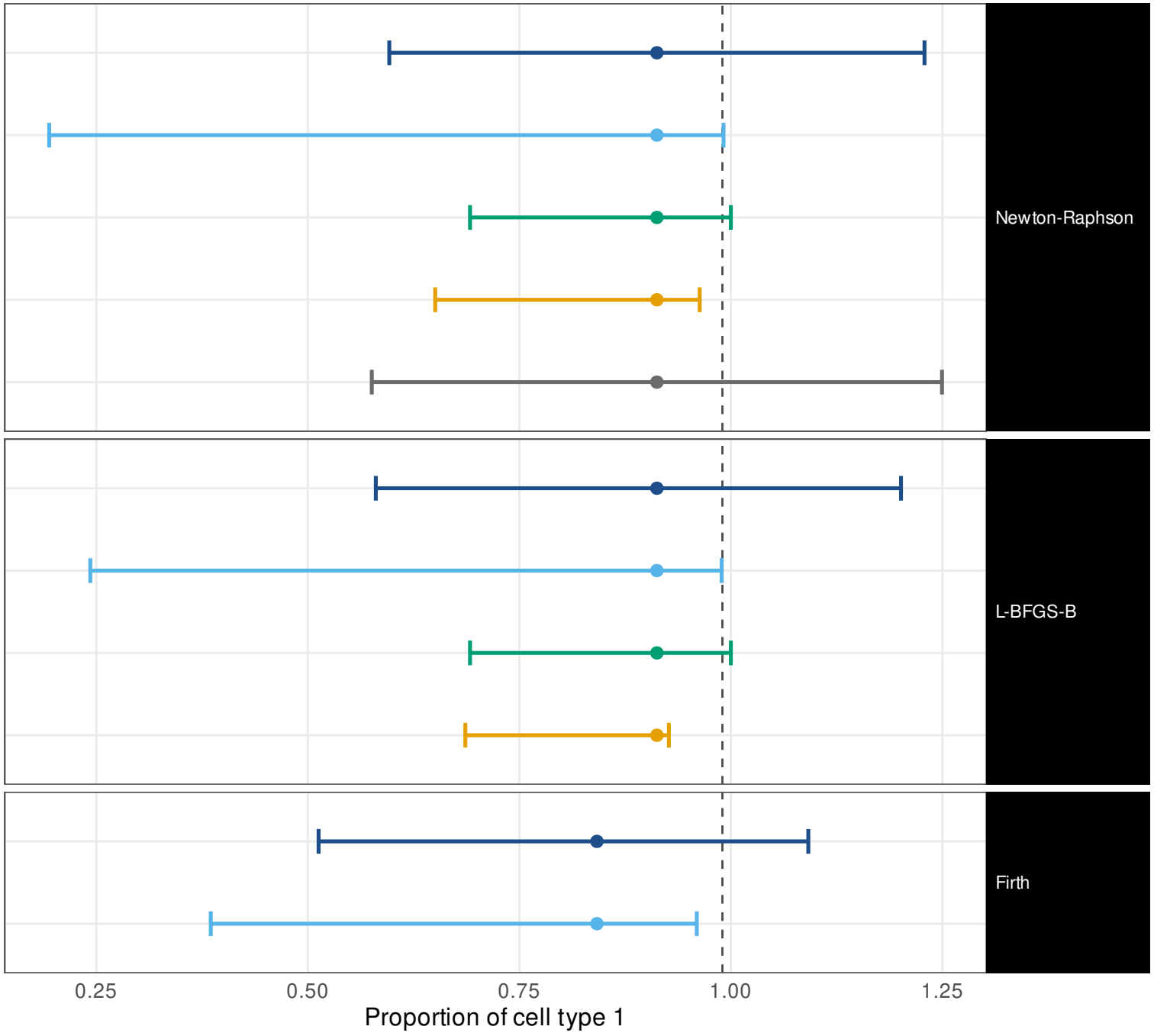
Confidence intervals: coverage [Wilson 95% interval] and mean width w

small centroid gap / heteroscedastic / highly unbalanced / rho = (+0.8, +0.8)

```
Wald (delta)              1  [0.57,    1]   w = 0.633
ILR, mapped back        0.6  [0.23, 0.88]   w = 0.797
Profile likelihood        1  [0.57,    1]   w = 0.308
Parametric bootstrap    0.4  [0.12, 0.77]   w = 0.313
Oracle (Fisher at p*)     1  [0.57,    1]   w = 0.674

Wald (delta)              1  [0.51,    1]   w = 0.621
ILR, mapped back        0.5  [0.15, 0.85]   w = 0.746
Profile likelihood        1  [0.57,    1]   w = 0.308
Parametric bootstrap    0.6  [0.23, 0.88]   w = 0.241

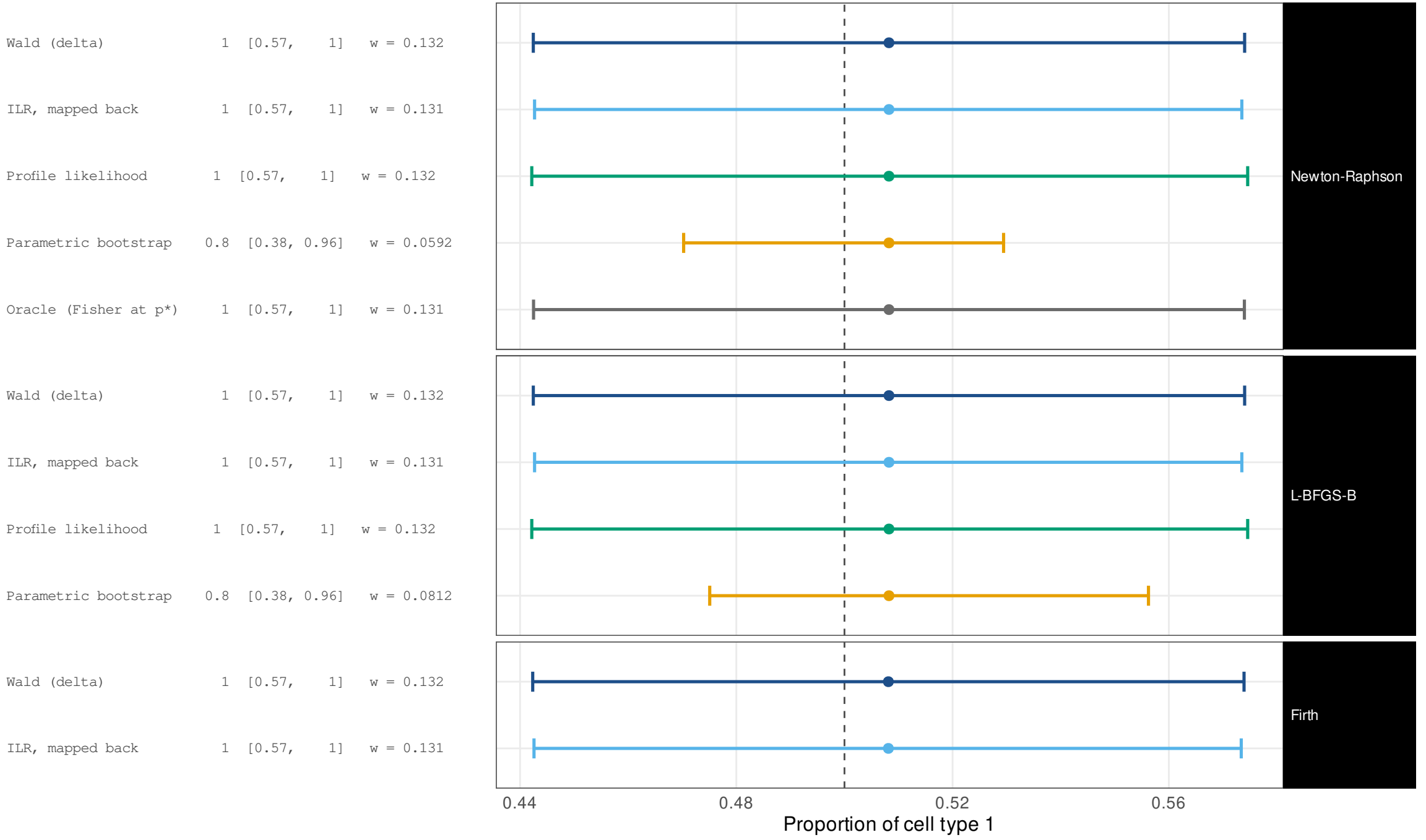
Wald (delta)              1  [0.51,    1]   w = 0.579
ILR, mapped back          0  [   0, 0.49]   w = 0.574
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

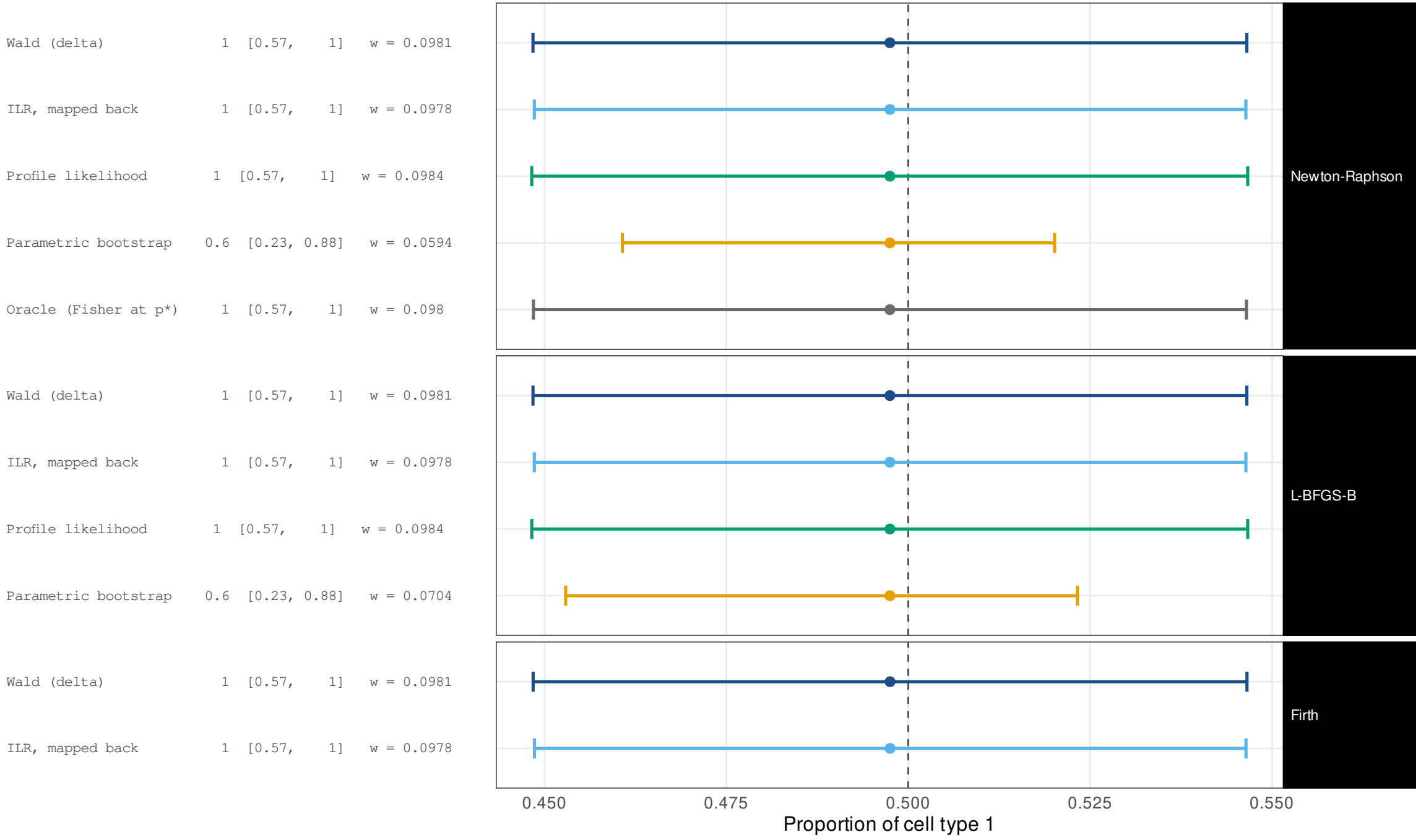
large centroid gap / homoscedastic / balanced / rho = (-0.8, -0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

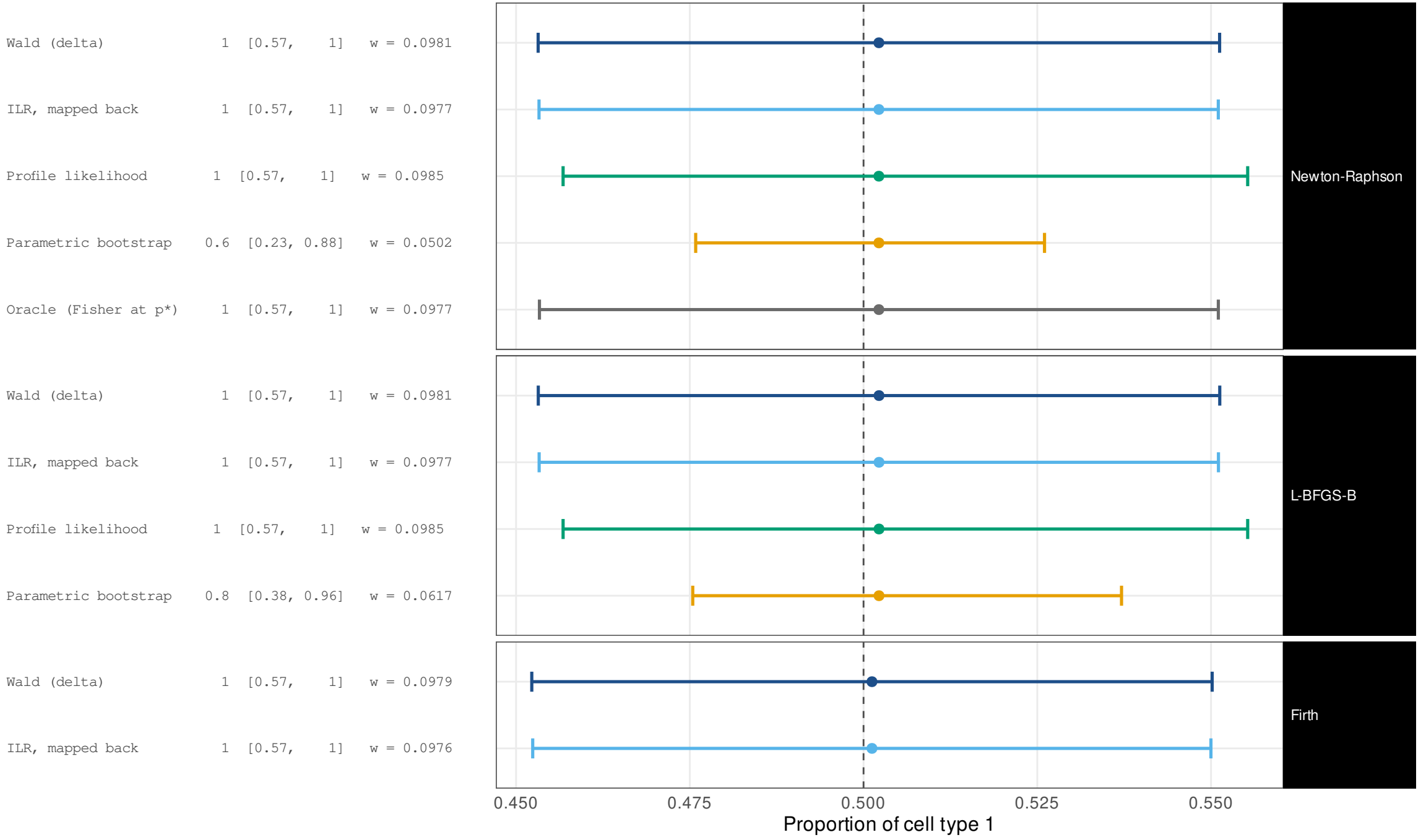
large centroid gap / homoscedastic / balanced / rho = (+0.0, +0.0)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

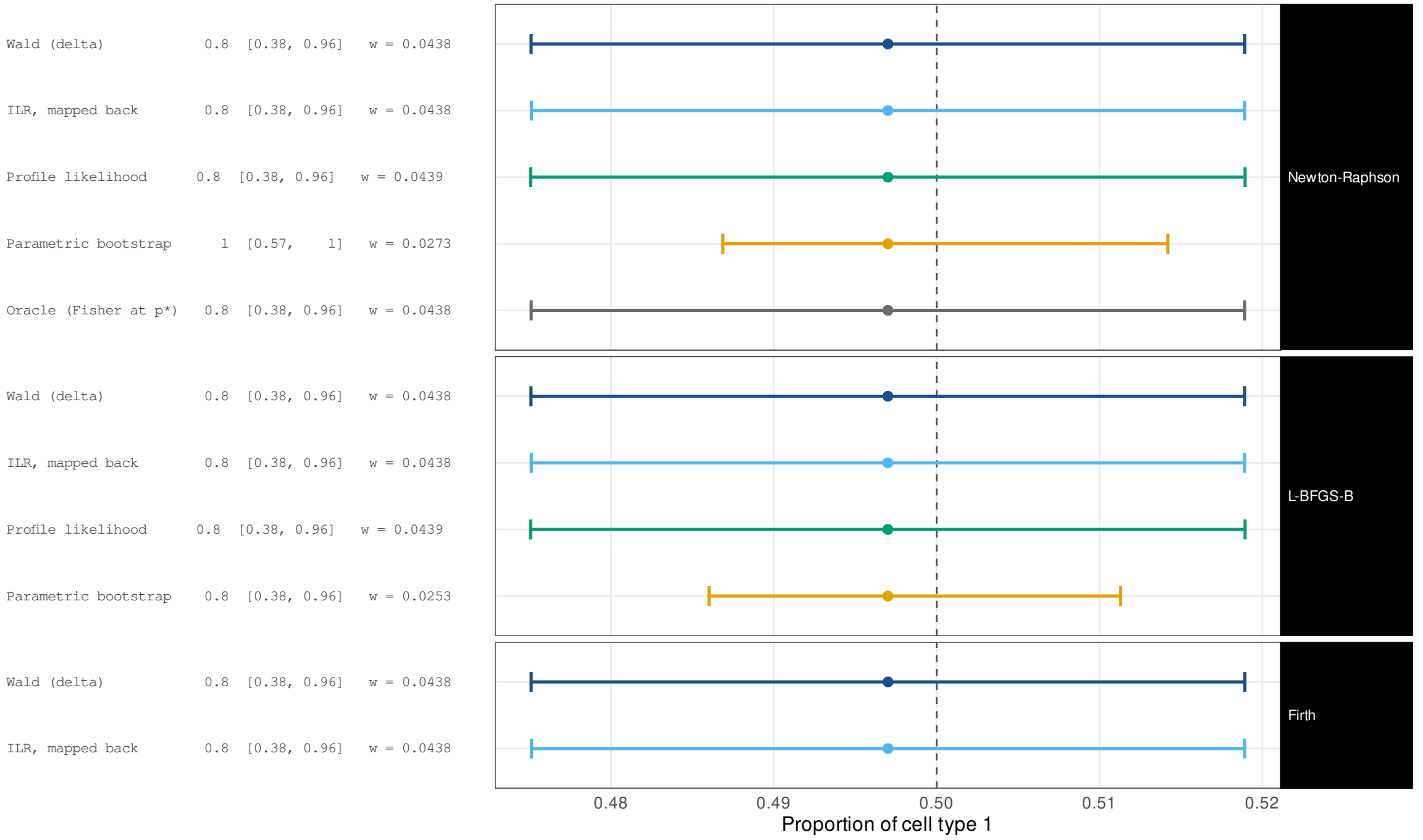
large centroid gap / homoscedastic / balanced / rho = (-0.8, +0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

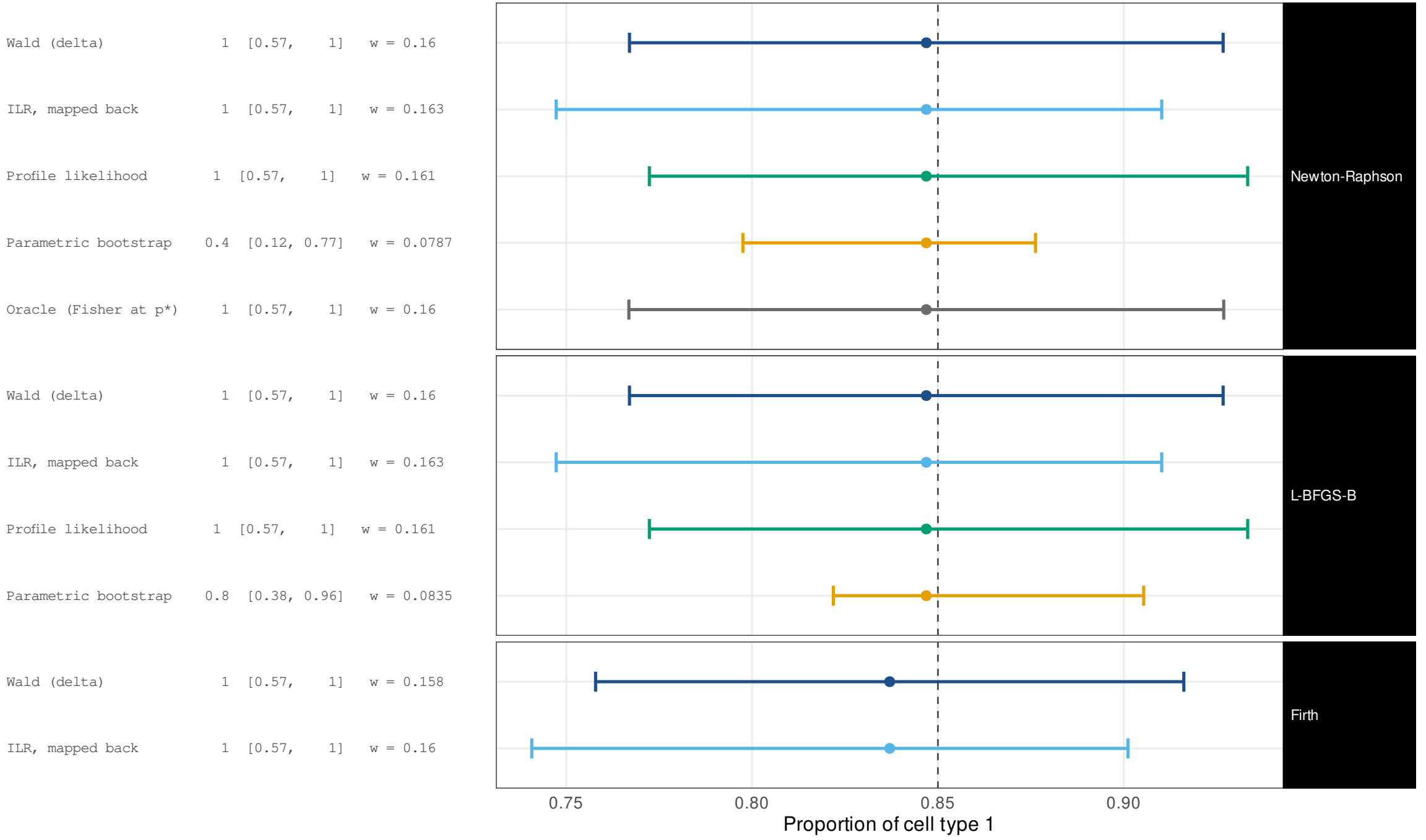
large centroid gap / homoscedastic / balanced / rho = (+0.8, +0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / homoscedastic / moderately unbalanced / rho = (-0.8, -0.8)

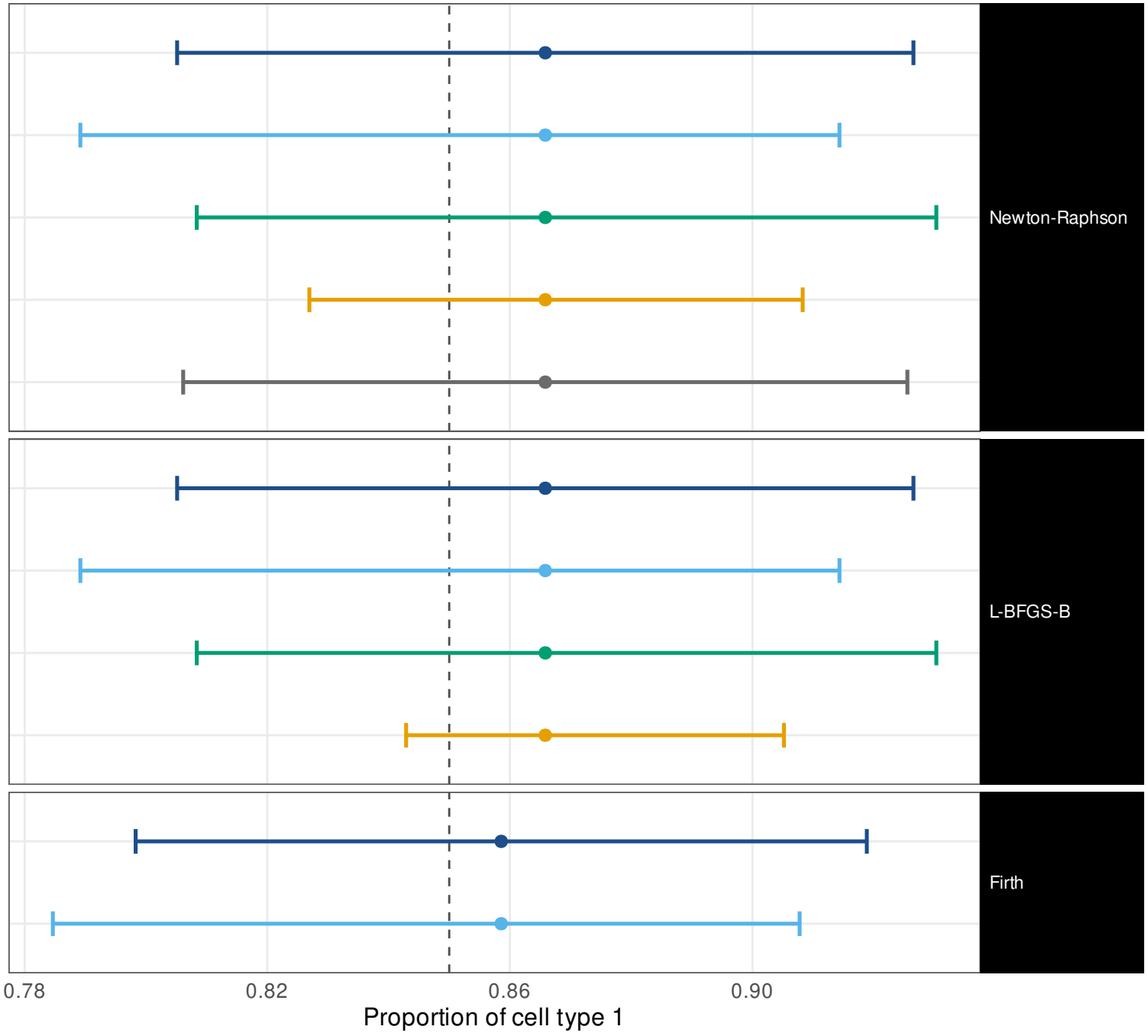


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / homoscedastic / moderately unbalanced / rho = (+0.0, +0.0)

Wald (delta)	0.8	[0.38, 0.96]	w = 0.121
ILR, mapped back	1	[0.57, 1]	w = 0.125
Profile likelihood	0.8	[0.38, 0.96]	w = 0.122
Parametric bootstrap	0.6	[0.23, 0.88]	w = 0.0813
Oracle (Fisher at p*)	0.8	[0.38, 0.96]	w = 0.119
Wald (delta)	0.8	[0.38, 0.96]	w = 0.121
ILR, mapped back	1	[0.57, 1]	w = 0.125
Profile likelihood	0.8	[0.38, 0.96]	w = 0.122
Parametric bootstrap	0.4	[0.12, 0.77]	w = 0.0623
Wald (delta)	0.8	[0.38, 0.96]	w = 0.121
ILR, mapped back	1	[0.57, 1]	w = 0.123



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

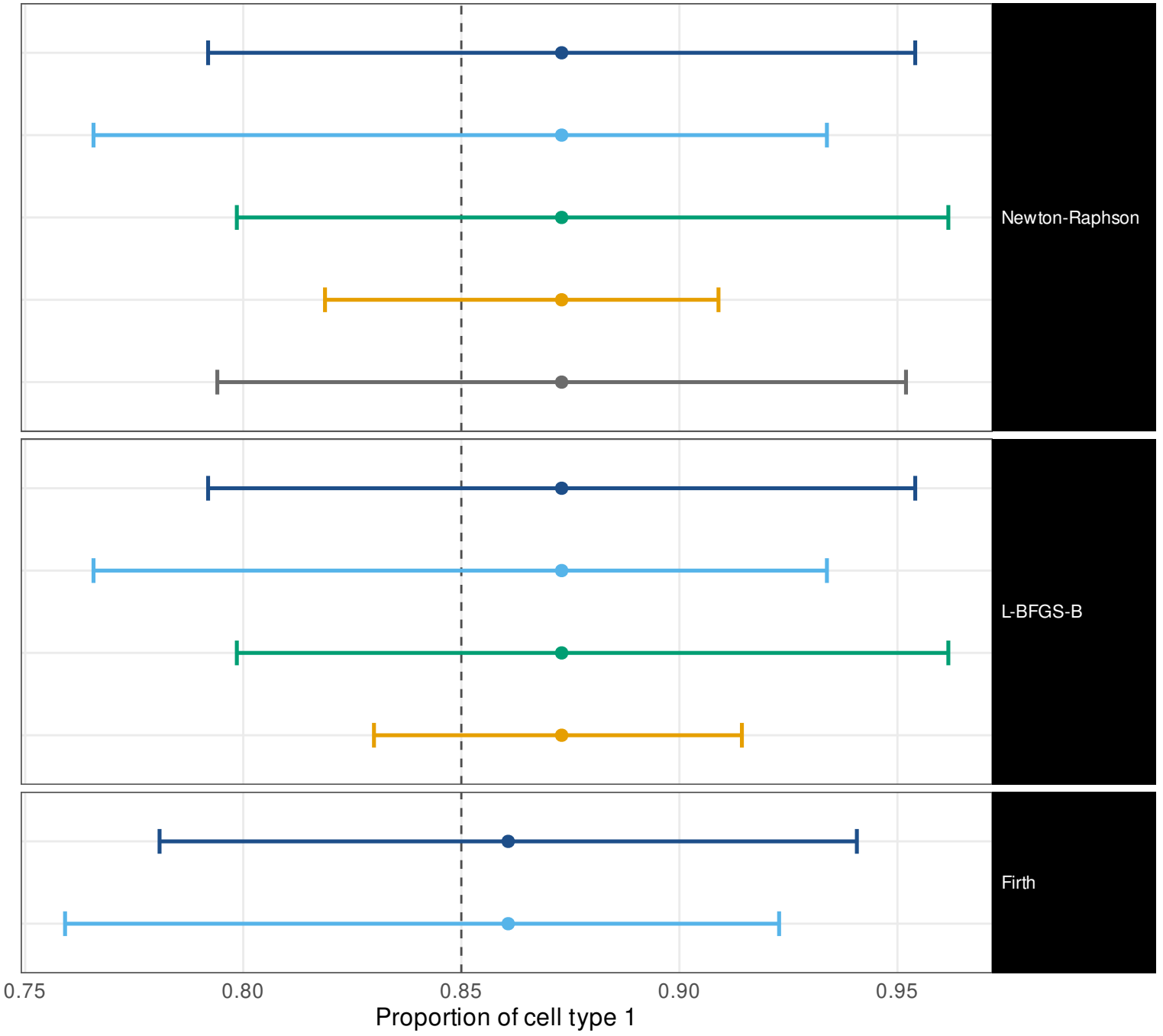
Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / homoscedastic / moderately unbalanced / rho = (-0.8, +0.8)

```
Wald (delta)              1   [0.57,     1]   w = 0.162
ILR, mapped back          1   [0.57,     1]   w = 0.168
Profile likelihood        1   [0.57,     1]   w = 0.163
Parametric bootstrap    0.8   [0.38,  0.96]   w = 0.0902
Oracle (Fisher at p*)     1   [0.57,     1]   w = 0.158

Wald (delta)              1   [0.57,     1]   w = 0.162
ILR, mapped back          1   [0.57,     1]   w = 0.168
Profile likelihood        1   [0.57,     1]   w = 0.163
Parametric bootstrap    0.4   [0.12,  0.77]   w = 0.0844

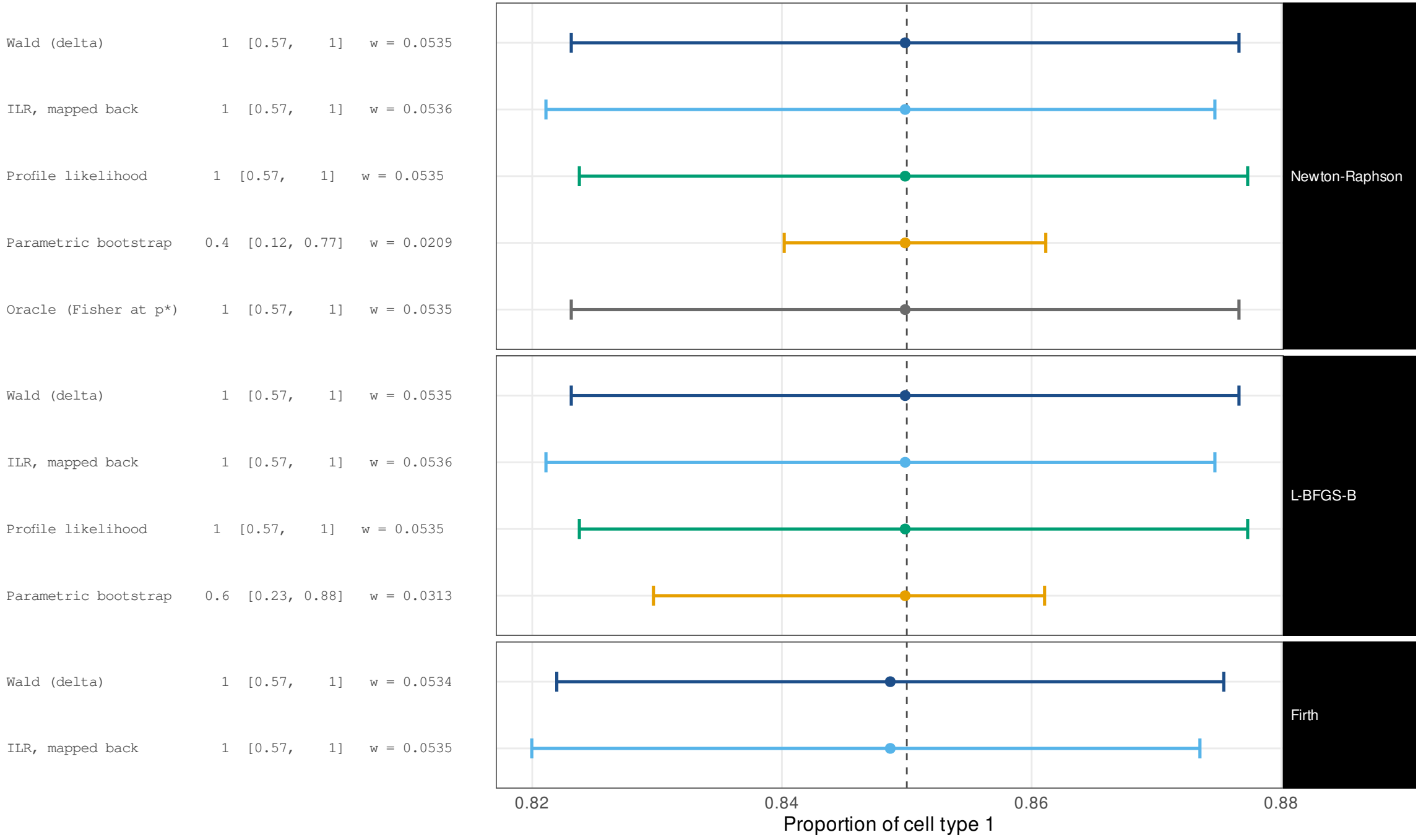
Wald (delta)              1   [0.57,     1]   w = 0.16
ILR, mapped back          1   [0.57,     1]   w = 0.164
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / homoscedastic / moderately unbalanced / rho = (+0.8, +0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

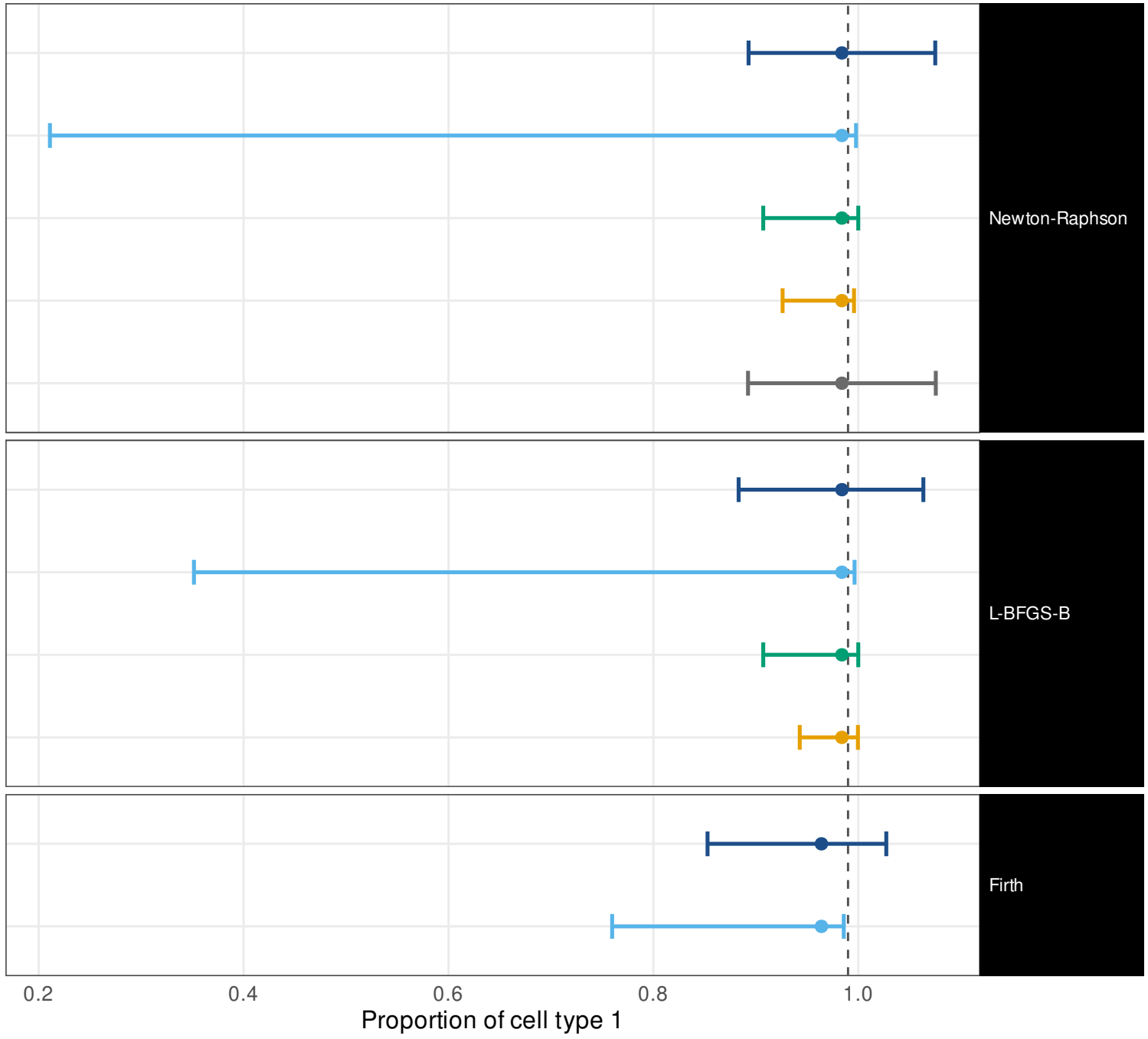
Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / homoscedastic / highly unbalanced / rho = (-0.8, -0.8)

```
Wald (delta)              1  [0.57,    1]   w = 0.182
ILR, mapped back        0.8  [0.38, 0.96]   w = 0.787
Profile likelihood        1  [0.57,    1]   w = 0.0928
Parametric bootstrap    0.6  [0.23, 0.88]   w = 0.0697
Oracle (Fisher at p*)     1  [0.57,    1]   w = 0.183

Wald (delta)              1  [0.44,    1]   w = 0.18
ILR, mapped back       0.67  [0.21, 0.94]   w = 0.645
Profile likelihood        1  [0.57,    1]   w = 0.0928
Parametric bootstrap      1  [0.57,    1]   w = 0.0568

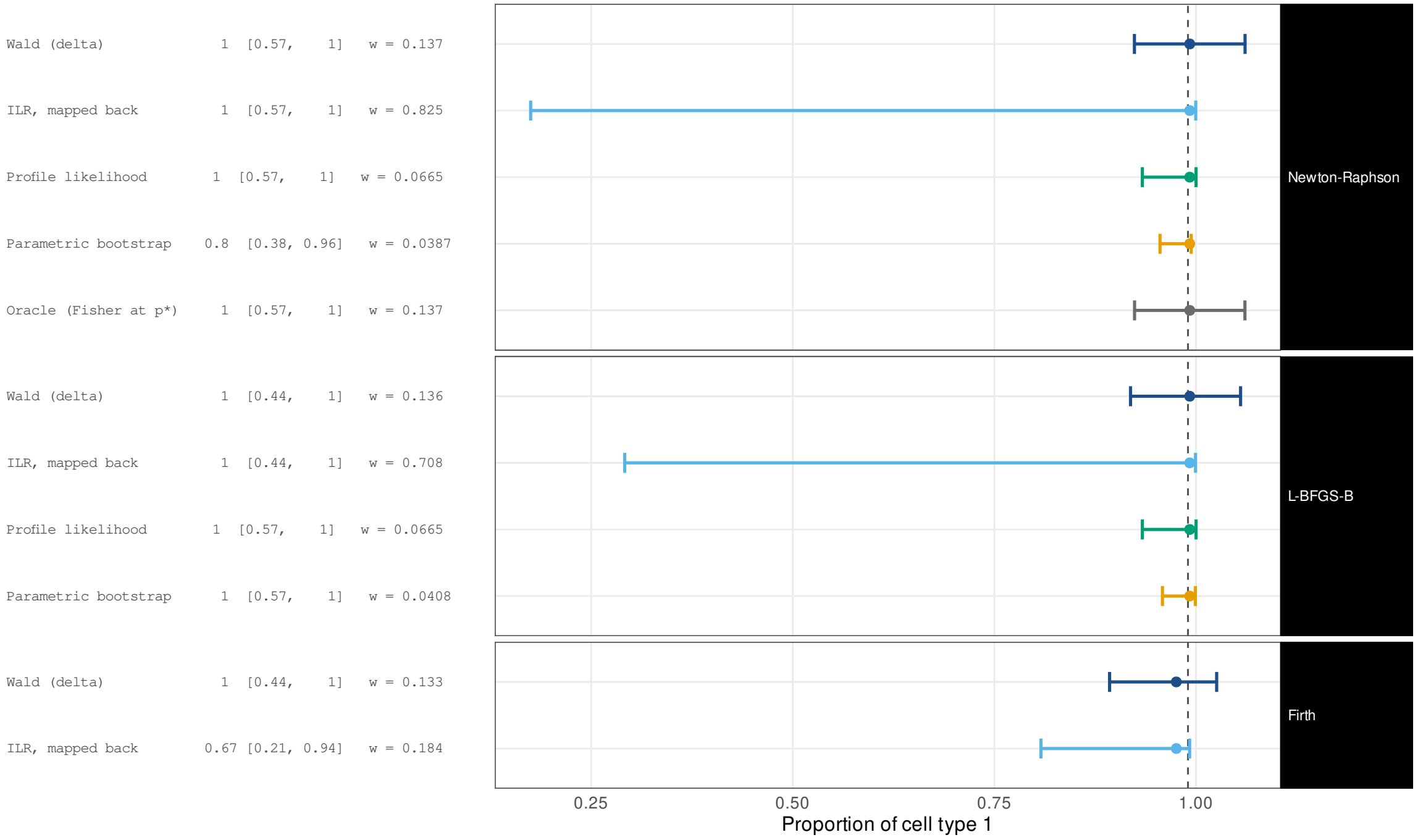
Wald (delta)              1  [0.44,    1]   w = 0.174
ILR, mapped back       0.33  [0.061, 0.79]   w = 0.226
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / homoscedastic / highly unbalanced / rho = (+0.0, +0.0)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

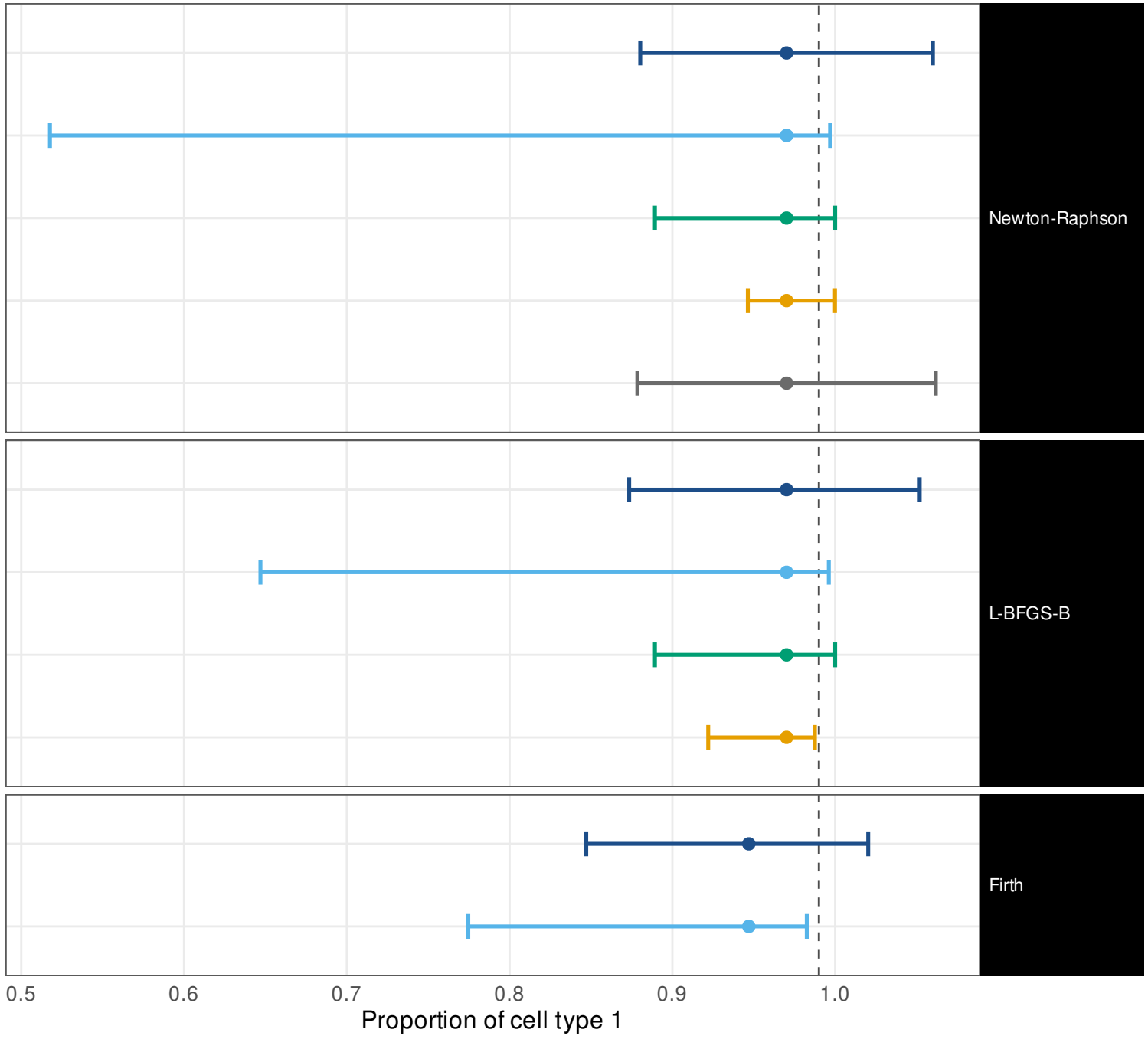
Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / homoscedastic / highly unbalanced / rho = (-0.8, +0.8)

```
Wald (delta)              1  [0.57,    1]   w = 0.18
ILR, mapped back          1  [0.57,    1]   w = 0.479
Profile likelihood        1  [0.57,    1]   w = 0.111
Parametric bootstrap      1  [0.57,    1]   w = 0.0535
Oracle (Fisher at p*)     1  [0.57,    1]   w = 0.183

Wald (delta)              1  [0.51,    1]   w = 0.178
ILR, mapped back          1  [0.51,    1]   w = 0.349
Profile likelihood        1  [0.57,    1]   w = 0.111
Parametric bootstrap    0.6  [0.23, 0.88]   w = 0.0655

Wald (delta)              1  [0.51,    1]   w = 0.173
ILR, mapped back          0  [   0, 0.49]   w = 0.208
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

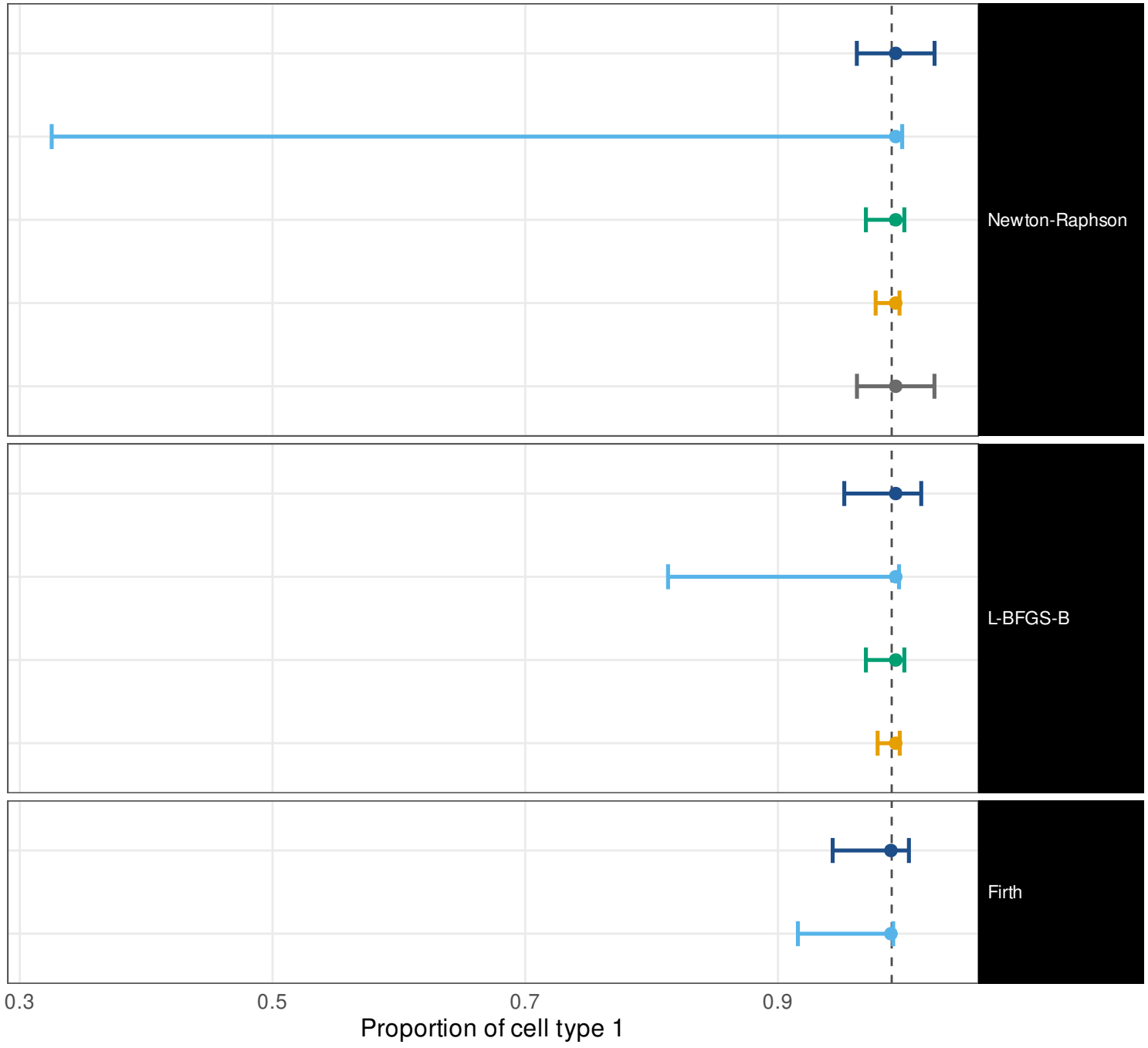
Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / homoscedastic / highly unbalanced / rho = (+0.8, +0.8)

```
Wald (delta)              1  [0.57,     1]   w = 0.0615
ILR, mapped back          1  [0.57,     1]   w = 0.673
Profile likelihood        1  [0.57,     1]   w = 0.0304
Parametric bootstrap    0.6  [0.23,  0.88]   w = 0.0189
Oracle (Fisher at p*)     1  [0.57,     1]   w = 0.0613

Wald (delta)              1  [0.34,     1]   w = 0.0609
ILR, mapped back          1  [0.34,     1]   w = 0.183
Profile likelihood        1  [0.57,     1]   w = 0.0304
Parametric bootstrap    0.8  [0.38,  0.96]   w = 0.0175

Wald (delta)              1  [0.34,     1]   w = 0.0603
ILR, mapped back        0.5  [0.095,  0.91]   w = 0.0754
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

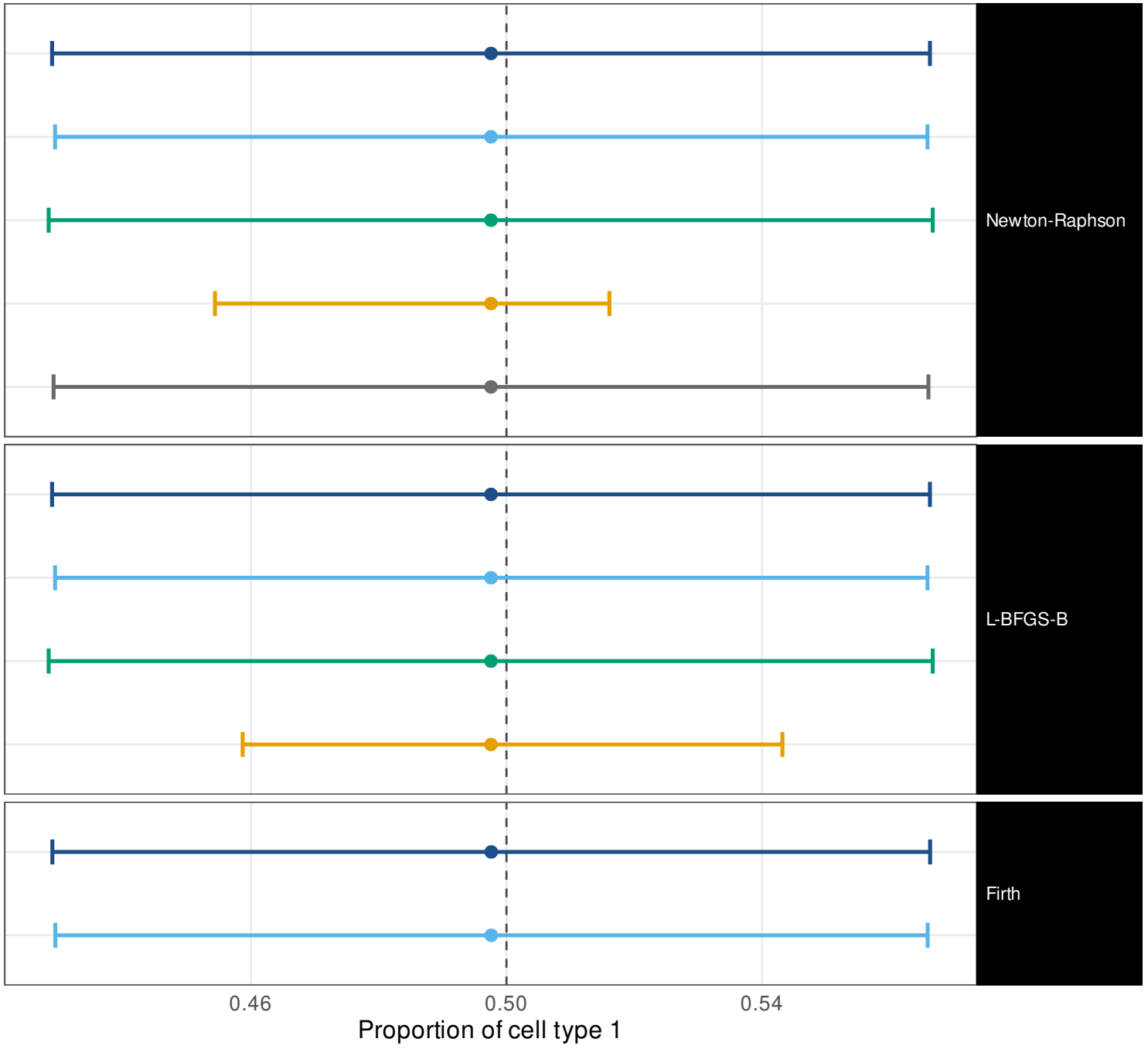
Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / heteroscedastic / balanced / rho = (-0.8, -0.8)

```
Wald (delta)              1  [0.57,     1]   w = 0.137
ILR, mapped back          1  [0.57,     1]   w = 0.137
Profile likelihood        1  [0.57,     1]   w = 0.138
Parametric bootstrap    0.2  [0.036, 0.62]   w = 0.0618
Oracle (Fisher at p*)     1  [0.57,     1]   w = 0.137

Wald (delta)              1  [0.57,     1]   w = 0.137
ILR, mapped back          1  [0.57,     1]   w = 0.137
Profile likelihood        1  [0.57,     1]   w = 0.138
Parametric bootstrap    0.8  [0.38,  0.96]   w = 0.0845

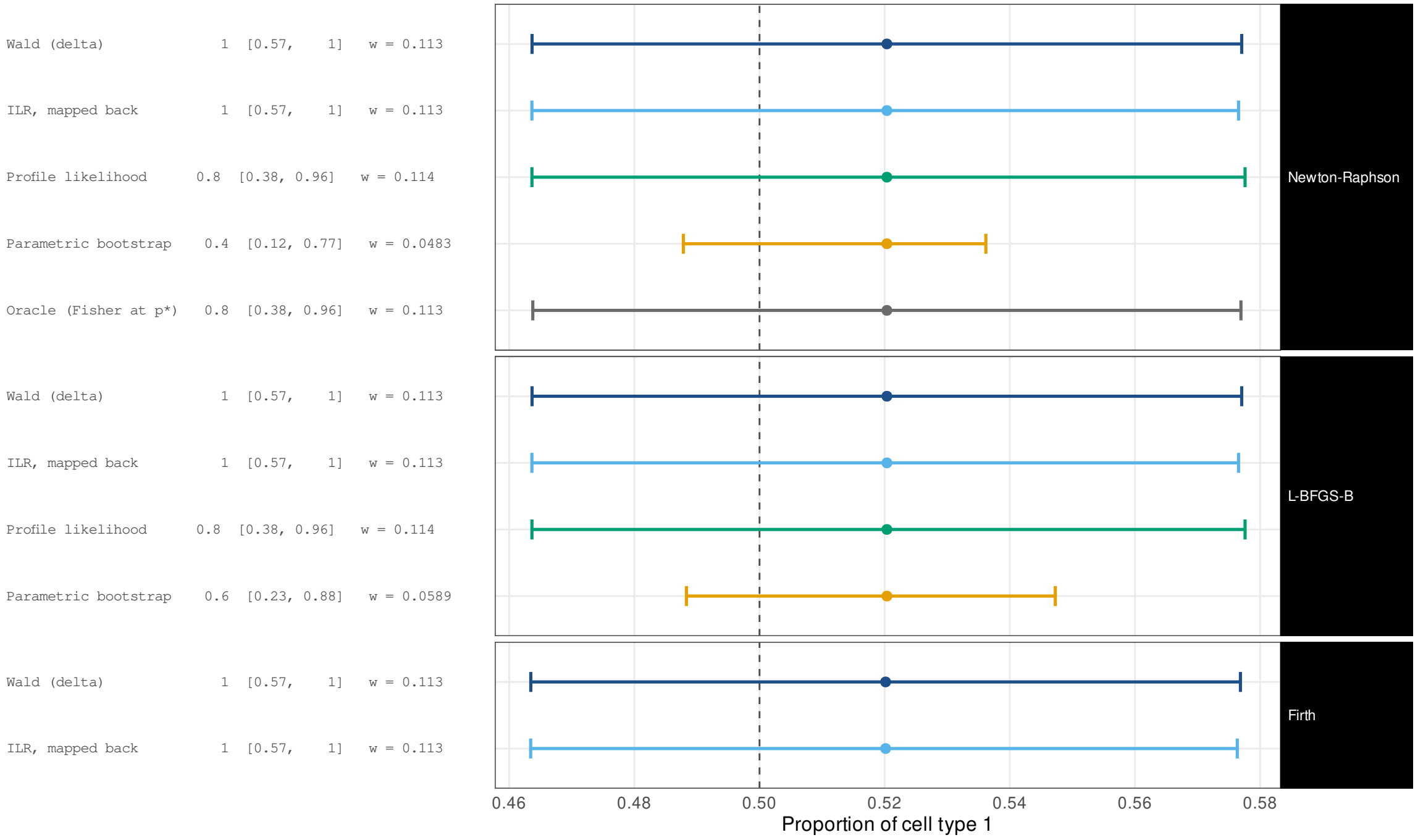
Wald (delta)              1  [0.57,     1]   w = 0.137
ILR, mapped back          1  [0.57,     1]   w = 0.137
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

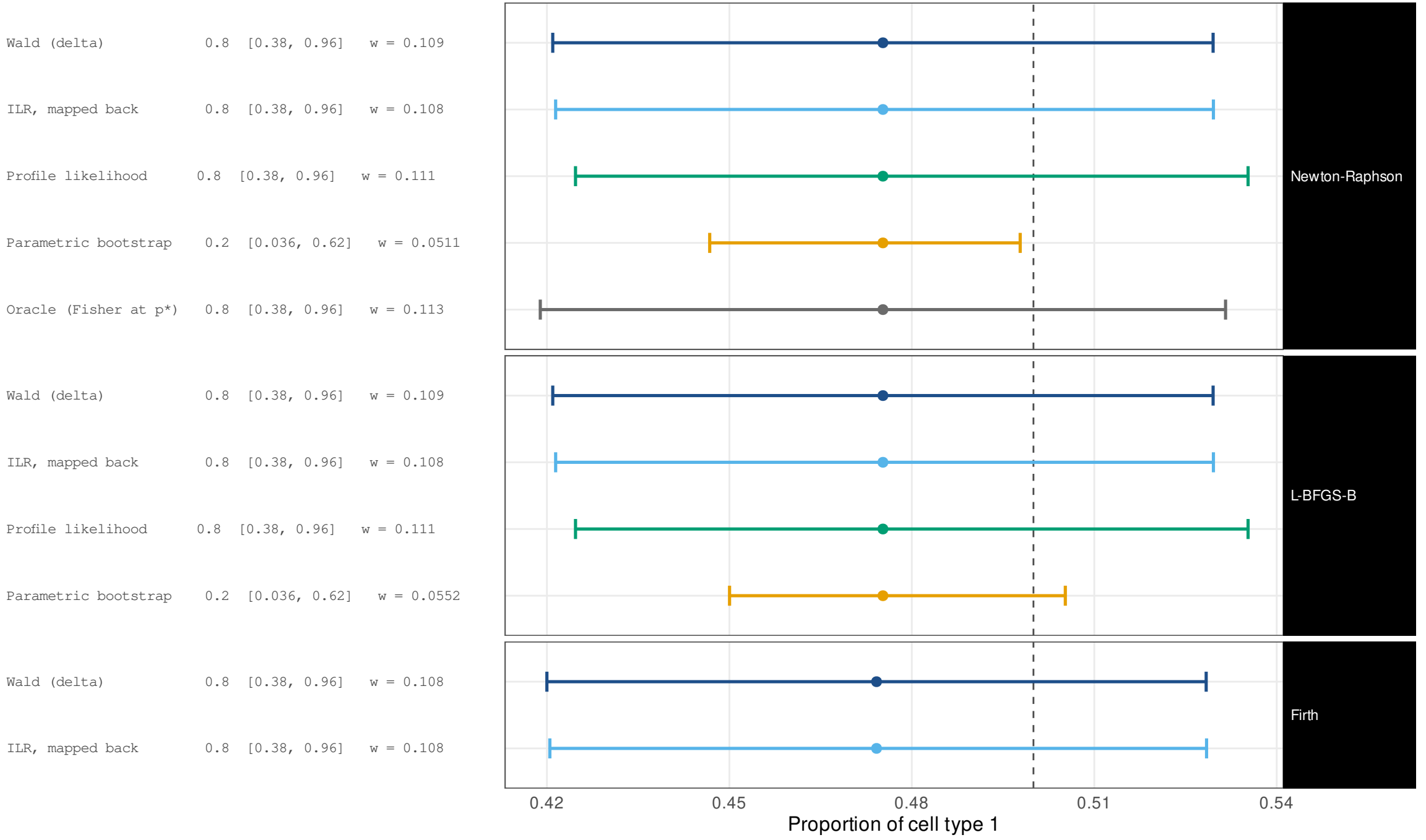
large centroid gap / heteroscedastic / balanced / rho = (+0.0, +0.0)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

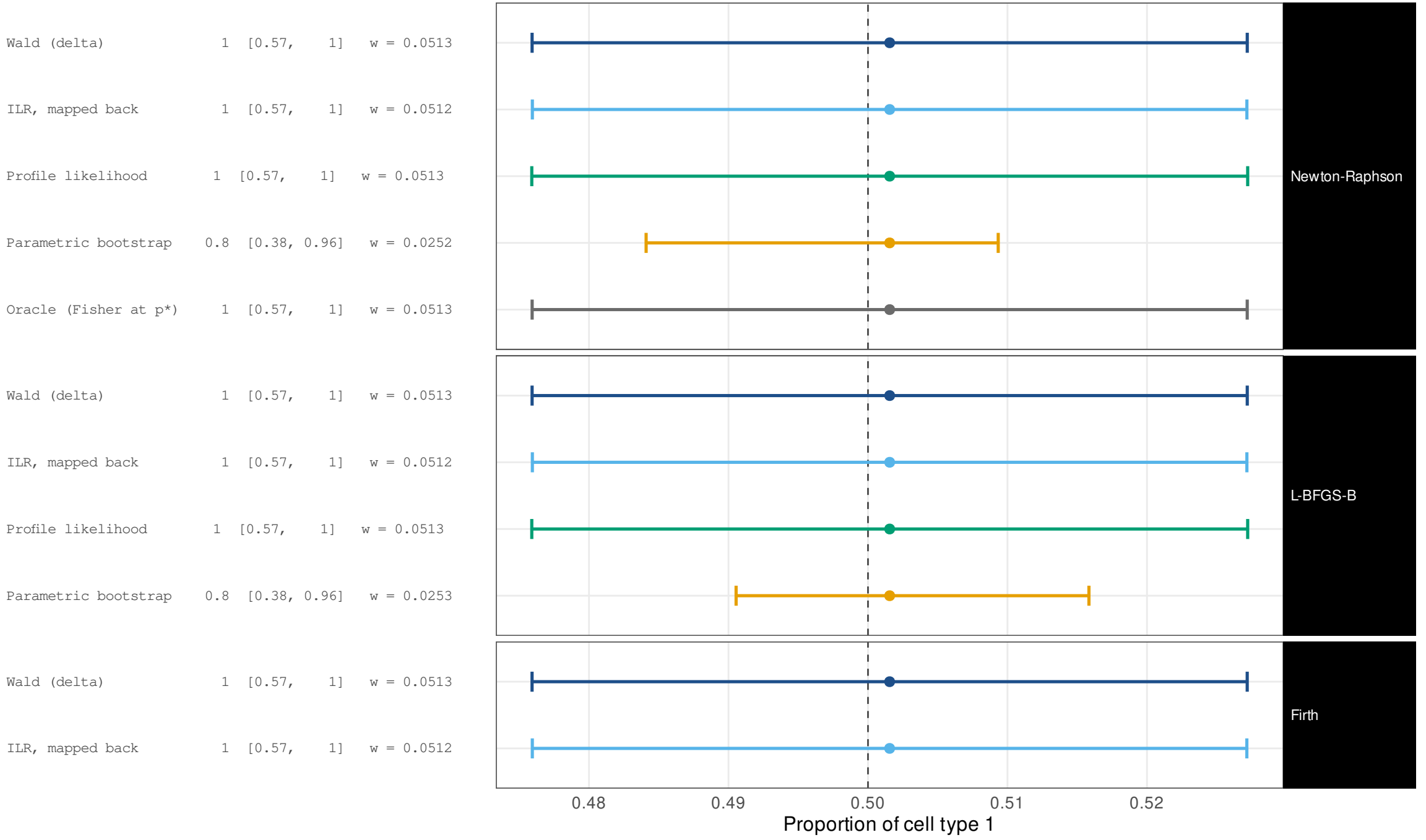
large centroid gap / heteroscedastic / balanced / rho = (-0.8, +0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / heteroscedastic / balanced / rho = (+0.8, +0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

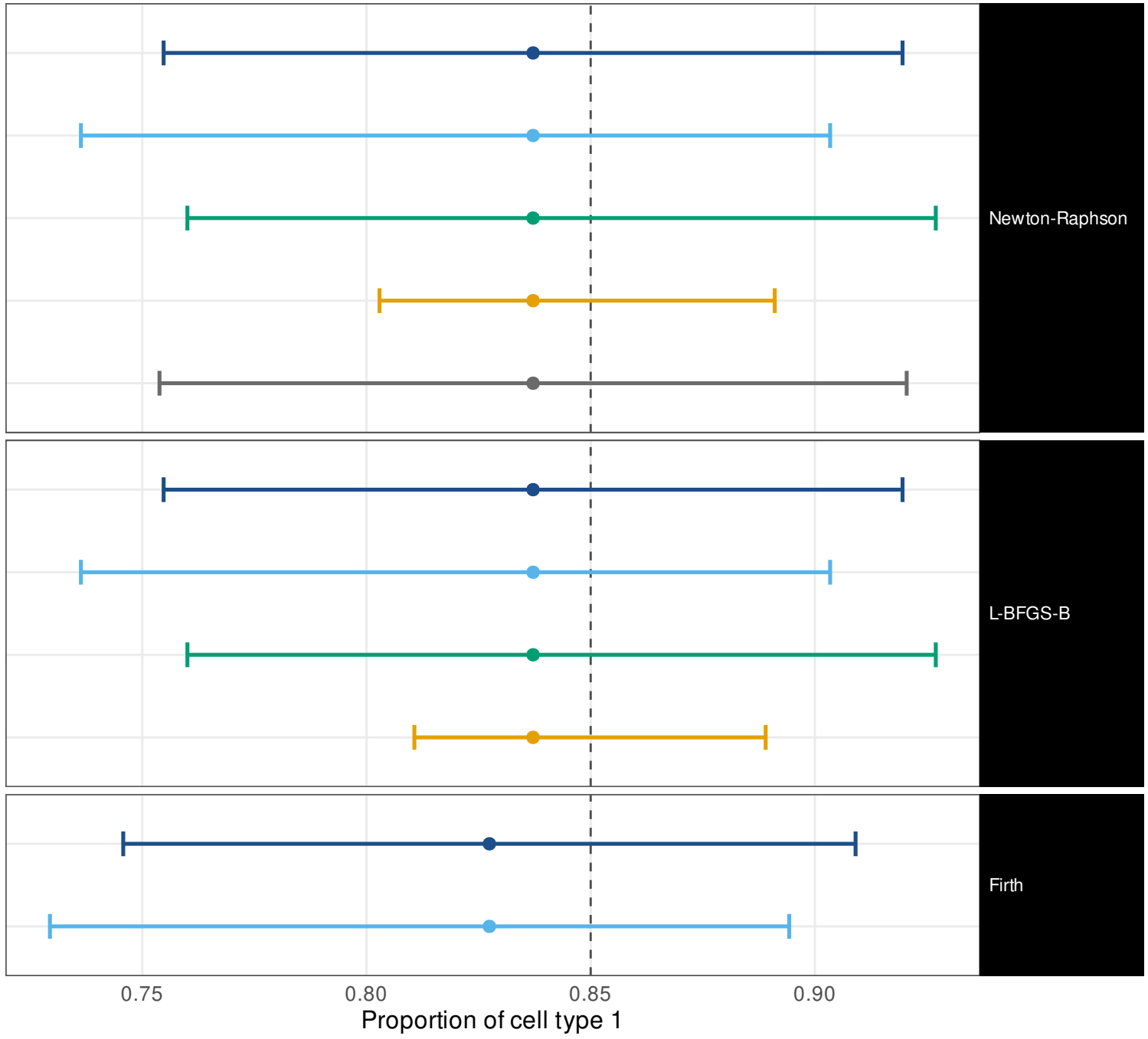
Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / heteroscedastic / moderately unbalanced / rho = (-0.8, -0.8)

```
Wald (delta)              1  [0.57,    1]   w = 0.165
ILR, mapped back          1  [0.57,    1]   w = 0.167
Profile likelihood        1  [0.57,    1]   w = 0.167
Parametric bootstrap      1  [0.57,    1]   w = 0.0881
Oracle (Fisher at p*)     1  [0.57,    1]   w = 0.167

Wald (delta)              1  [0.57,    1]   w = 0.165
ILR, mapped back          1  [0.57,    1]   w = 0.167
Profile likelihood        1  [0.57,    1]   w = 0.167
Parametric bootstrap    0.8  [0.38, 0.96]   w = 0.0783

Wald (delta)              1  [0.57,    1]   w = 0.163
ILR, mapped back          1  [0.57,    1]   w = 0.165
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

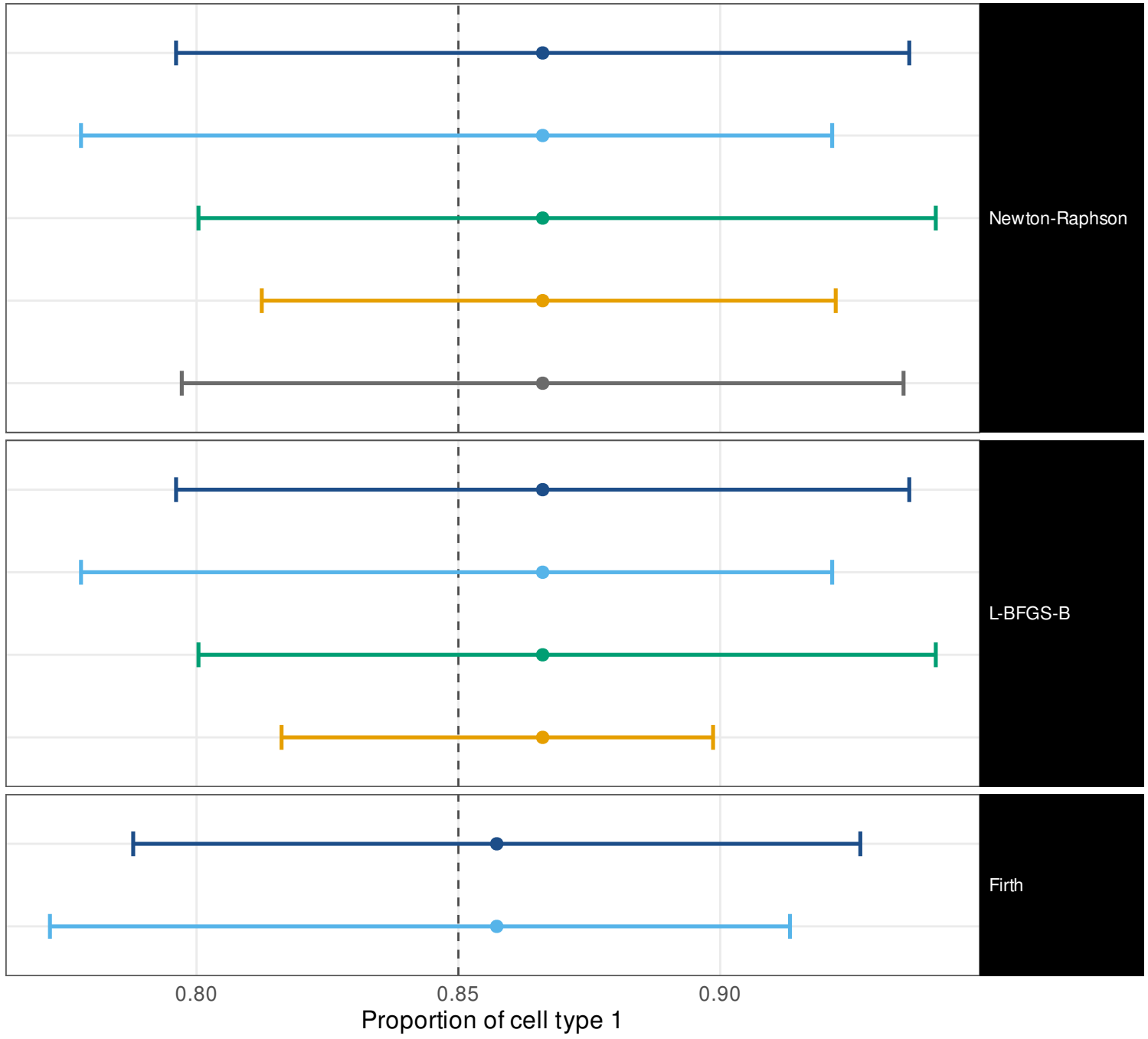
Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / heteroscedastic / moderately unbalanced / rho = (+0.0, +0.0)

```
Wald (delta)              1   [0.57,     1]   w = 0.14
ILR, mapped back          1   [0.57,     1]   w = 0.143
Profile likelihood        1   [0.57,     1]   w = 0.141
Parametric bootstrap    0.8   [0.38,  0.96]   w = 0.11
Oracle (Fisher at p*)     1   [0.57,     1]   w = 0.138

Wald (delta)              1   [0.57,     1]   w = 0.14
ILR, mapped back          1   [0.57,     1]   w = 0.143
Profile likelihood        1   [0.57,     1]   w = 0.141
Parametric bootstrap    0.8   [0.38,  0.96]   w = 0.0824

Wald (delta)              1   [0.57,     1]   w = 0.139
ILR, mapped back          1   [0.57,     1]   w = 0.141
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

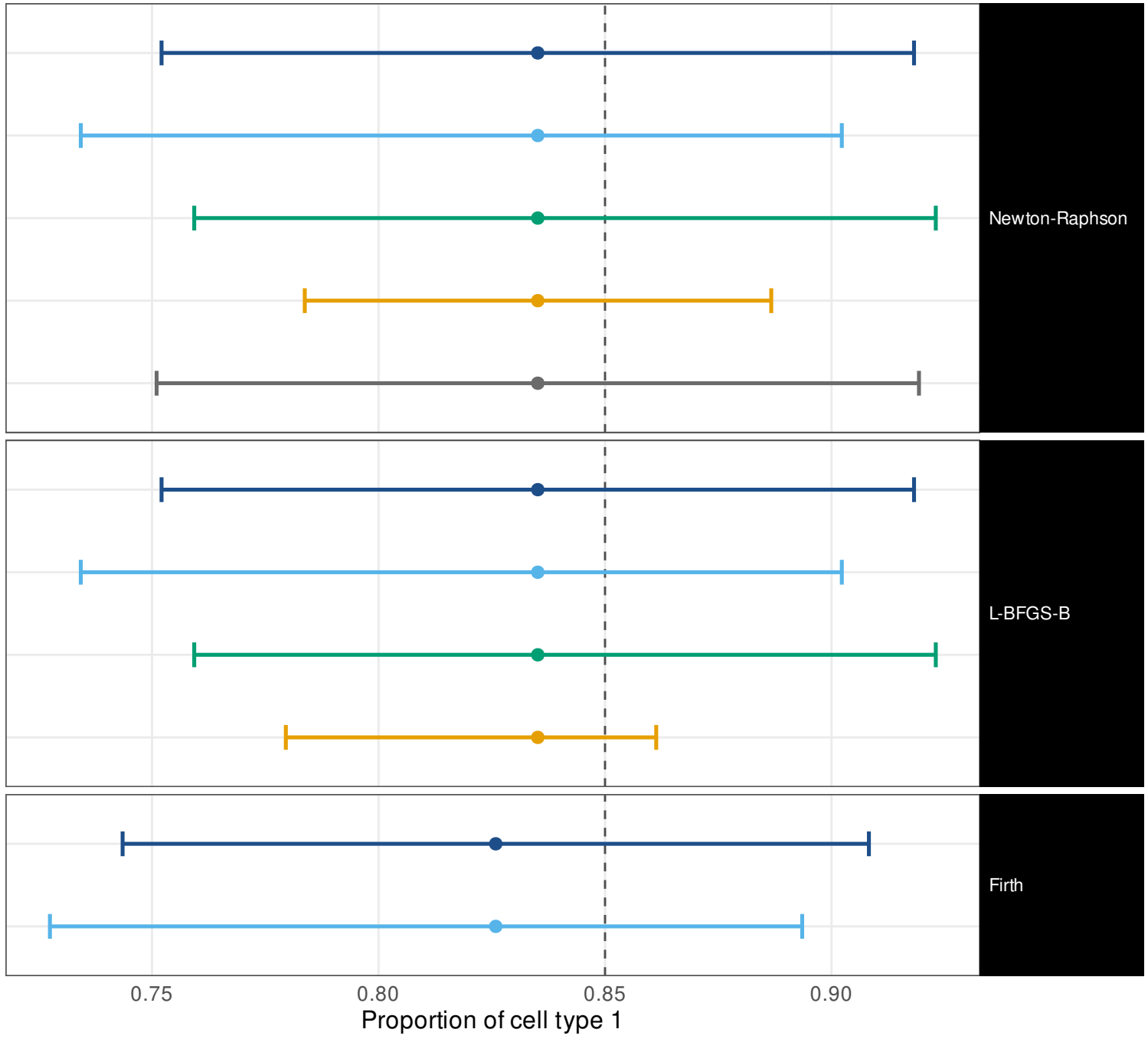
Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / heteroscedastic / moderately unbalanced / rho = (-0.8, +0.8)

```
Wald (delta)              1   [0.57,     1]   w = 0.166
ILR, mapped back          1   [0.57,     1]   w = 0.168
Profile likelihood        1   [0.57,     1]   w = 0.164
Parametric bootstrap    0.8   [0.38,  0.96]   w = 0.103
Oracle (Fisher at p*)     1   [0.57,     1]   w = 0.168

Wald (delta)              1   [0.57,     1]   w = 0.166
ILR, mapped back          1   [0.57,     1]   w = 0.168
Profile likelihood        1   [0.57,     1]   w = 0.164
Parametric bootstrap    0.8   [0.38,  0.96]   w = 0.0818

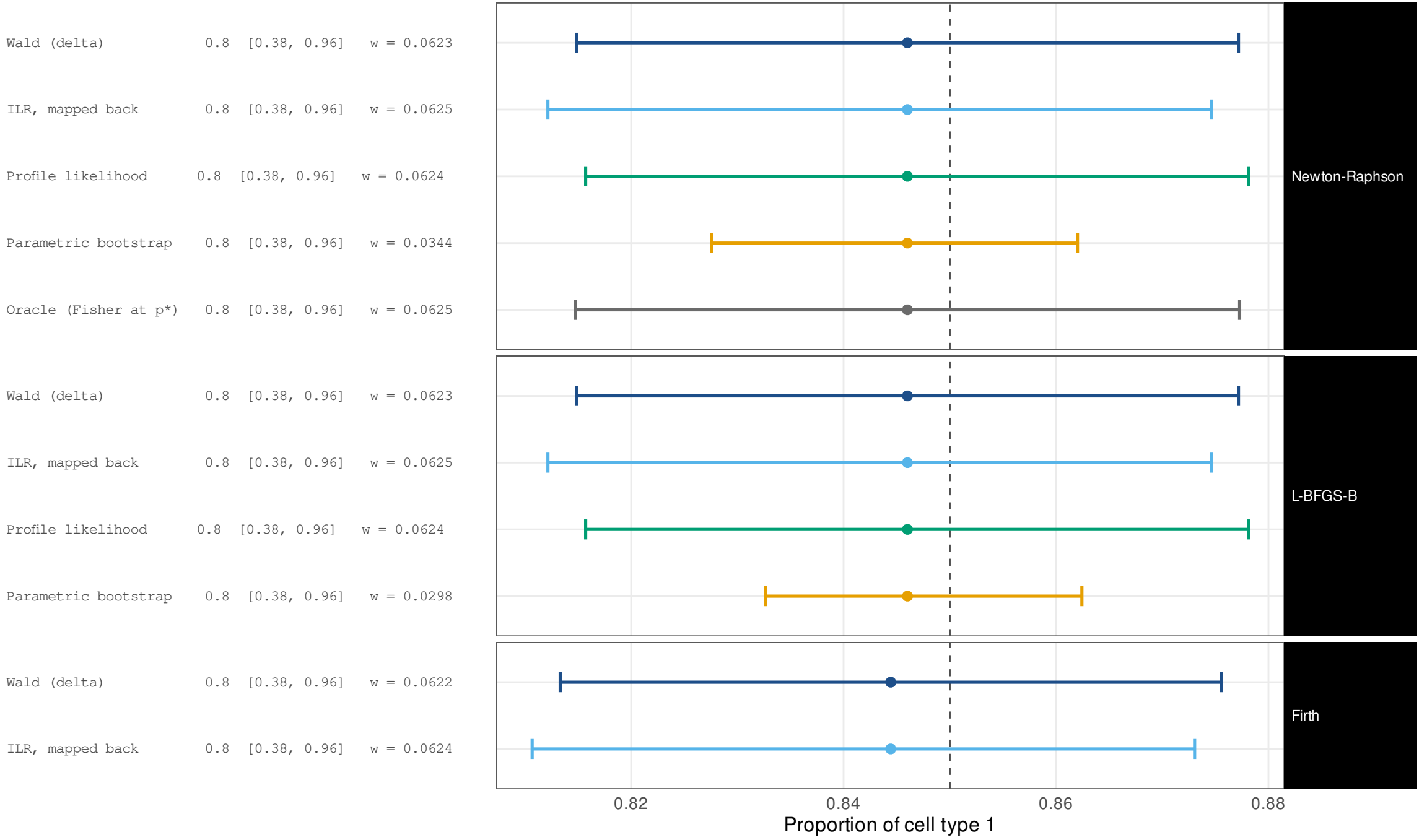
Wald (delta)              1   [0.57,     1]   w = 0.165
ILR, mapped back          1   [0.57,     1]   w = 0.166
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / heteroscedastic / moderately unbalanced / rho = (+0.8, +0.8)



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

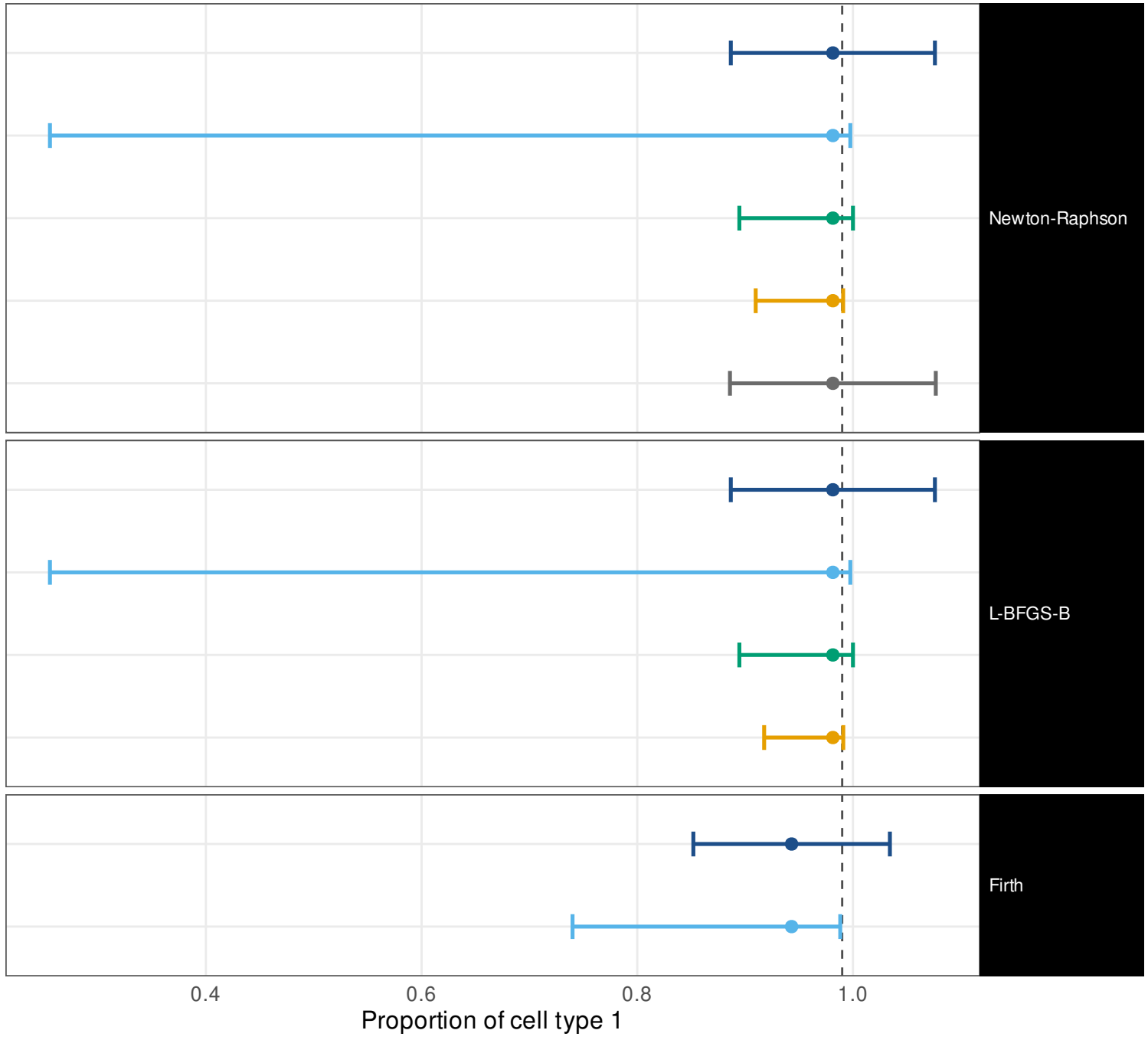
Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / heteroscedastic / highly unbalanced / rho = (-0.8, -0.8)

```
Wald (delta)              1  [0.57,    1]   w = 0.189
ILR, mapped back        0.8  [0.38, 0.96]   w = 0.742
Profile likelihood        1  [0.57,    1]   w = 0.105
Parametric bootstrap    0.8  [0.38, 0.96]   w = 0.0811
Oracle (Fisher at p*)     1  [0.57,    1]   w = 0.191

Wald (delta)              1  [0.57,    1]   w = 0.189
ILR, mapped back        0.8  [0.38, 0.96]   w = 0.742
Profile likelihood        1  [0.57,    1]   w = 0.105
Parametric bootstrap    0.8  [0.38, 0.96]   w = 0.0734

Wald (delta)              1  [0.57,    1]   w = 0.182
ILR, mapped back        0.6  [0.23, 0.88]   w = 0.248
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

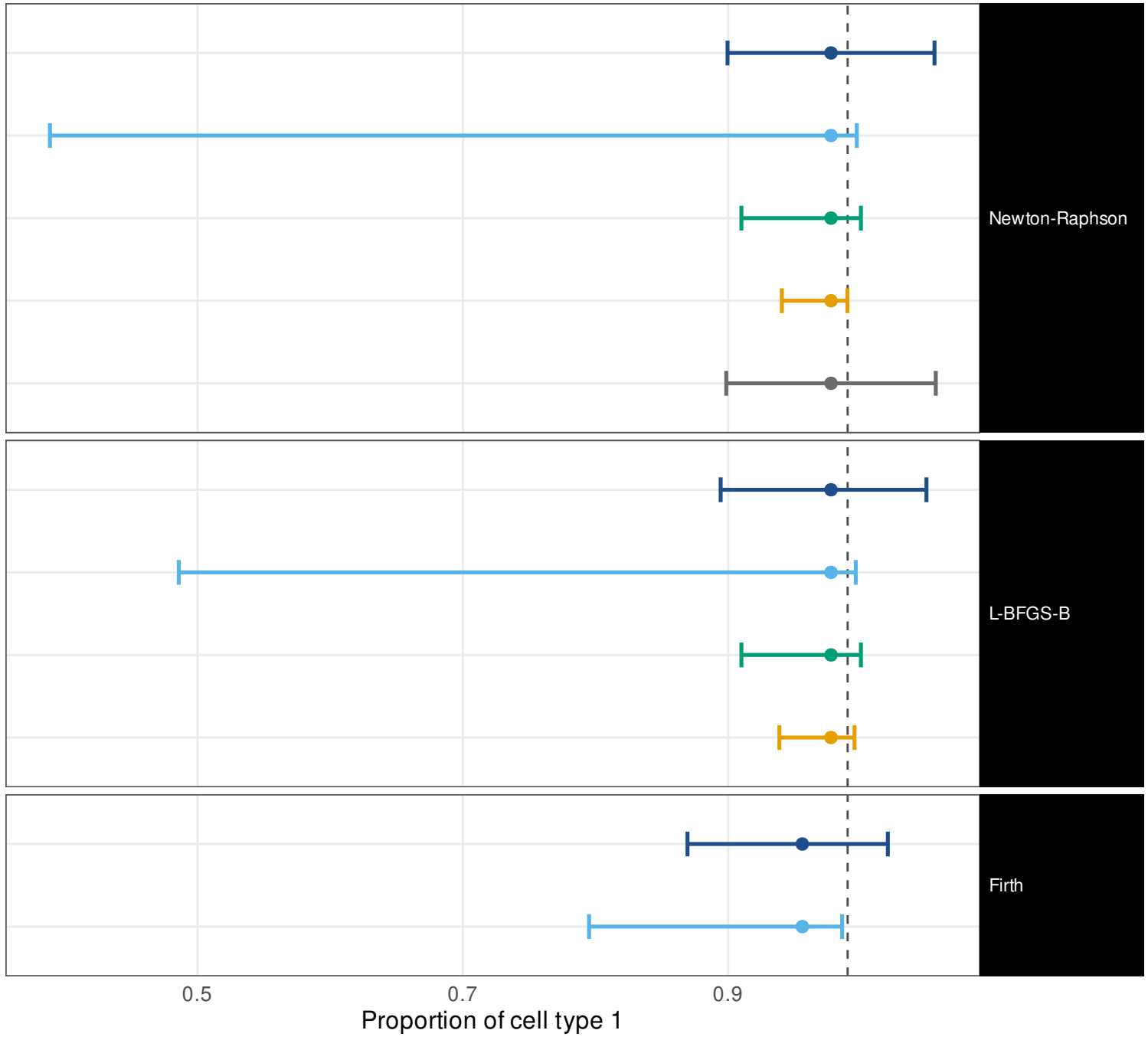
Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / heteroscedastic / highly unbalanced / rho = (+0.0, +0.0)

```
Wald (delta)              1  [0.57,    1]   w = 0.156
ILR, mapped back        0.8  [0.38, 0.96]   w = 0.608
Profile likelihood        1  [0.57,    1]   w = 0.0901
Parametric bootstrap    0.6  [0.23, 0.88]   w = 0.0494
Oracle (Fisher at p*)     1  [0.57,    1]   w = 0.158

Wald (delta)              1  [0.51,    1]   w = 0.155
ILR, mapped back       0.75  [ 0.3, 0.95]   w = 0.51
Profile likelihood        1  [0.57,    1]   w = 0.0901
Parametric bootstrap    0.8  [0.38, 0.96]   w = 0.0566

Wald (delta)              1  [0.51,    1]   w = 0.151
ILR, mapped back        0.5  [0.15, 0.85]   w = 0.191
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

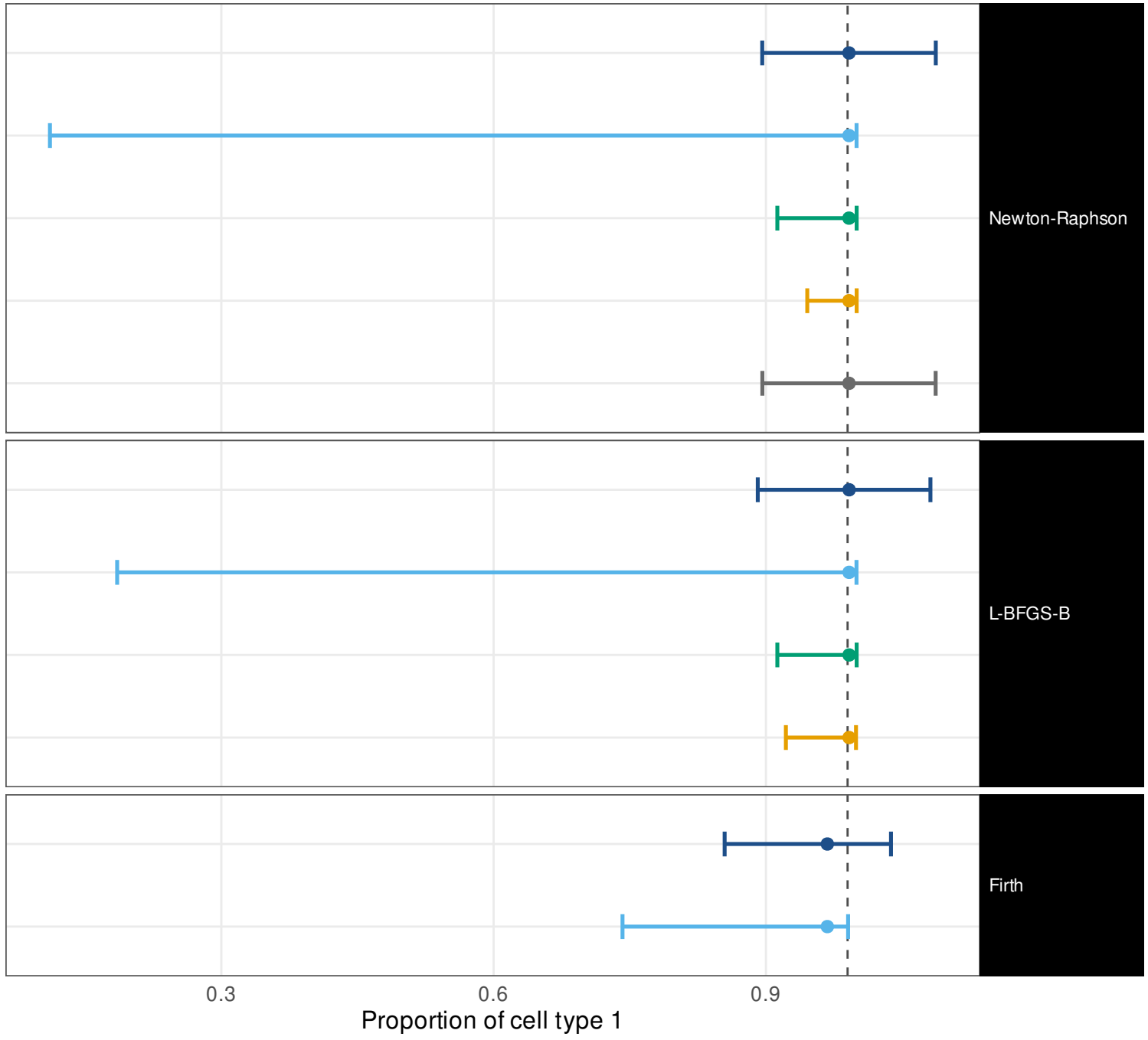
Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / heteroscedastic / highly unbalanced / rho = (-0.8, +0.8)

```
Wald (delta)              1  [0.57,    1]   w = 0.191
ILR, mapped back          1  [0.57,    1]   w = 0.889
Profile likelihood        1  [0.57,    1]   w = 0.0874
Parametric bootstrap      1  [0.57,    1]   w = 0.0544
Oracle (Fisher at p*)     1  [0.57,    1]   w = 0.191

Wald (delta)              1  [0.44,    1]   w = 0.19
ILR, mapped back          1  [0.44,    1]   w = 0.815
Profile likelihood        1  [0.57,    1]   w = 0.0874
Parametric bootstrap      1  [0.57,    1]   w = 0.0773

Wald (delta)              1  [0.44,    1]   w = 0.183
ILR, mapped back       0.67  [0.21, 0.94]   w = 0.249
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

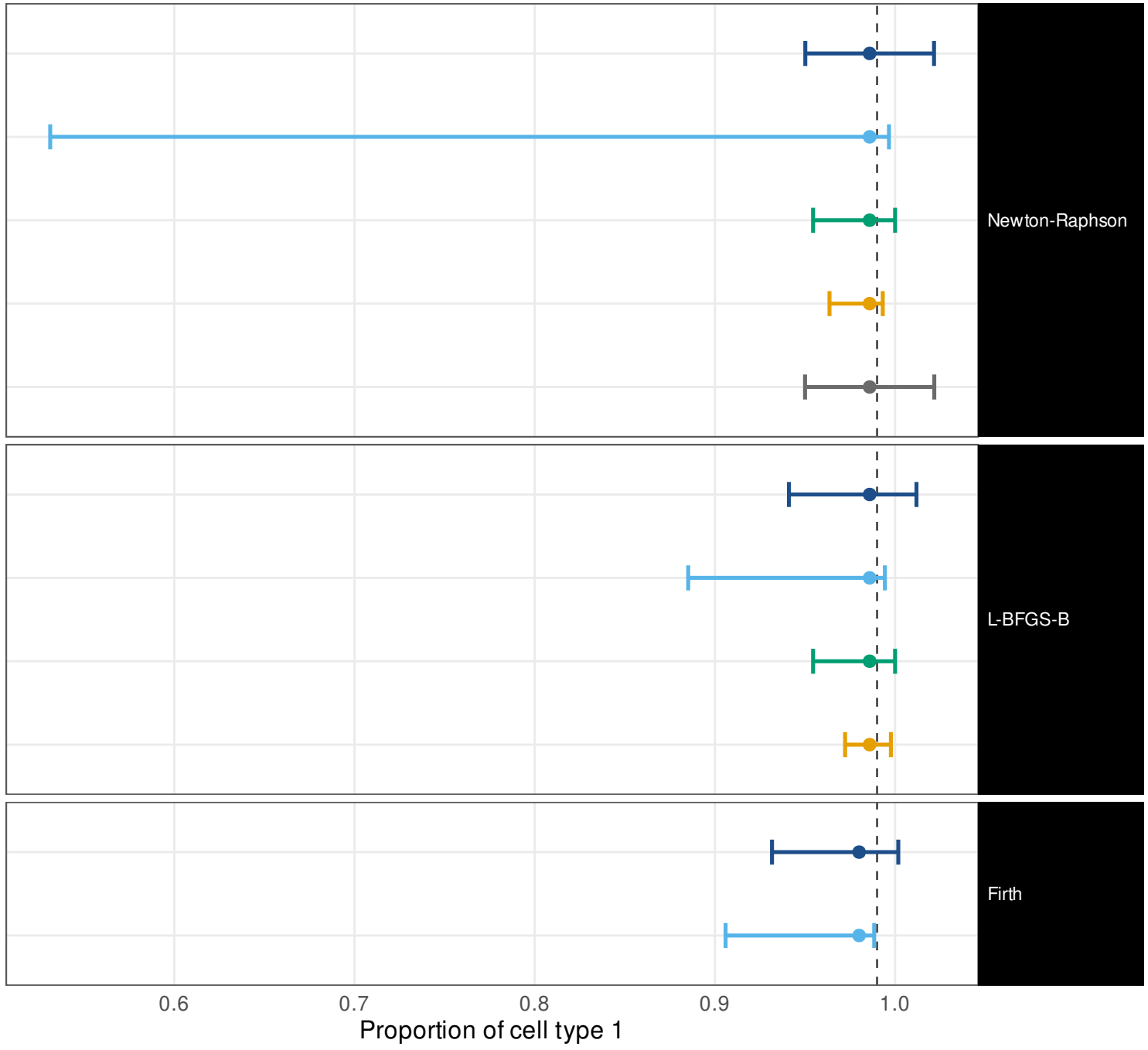
Confidence intervals: coverage [Wilson 95% interval] and mean width w

large centroid gap / heteroscedastic / highly unbalanced / rho = (+0.8, +0.8)

```
Wald (delta)              1  [0.57,     1]   w = 0.0714
ILR, mapped back          1  [0.57,     1]   w = 0.465
Profile likelihood        1  [0.57,     1]   w = 0.0455
Parametric bootstrap    0.8  [0.38,  0.96]   w = 0.0296
Oracle (Fisher at p*)     1  [0.57,     1]   w = 0.0717

Wald (delta)              1  [0.44,     1]   w = 0.0708
ILR, mapped back          1  [0.44,     1]   w = 0.109
Profile likelihood        1  [0.57,     1]   w = 0.0455
Parametric bootstrap    0.8  [0.38,  0.96]   w = 0.0254

Wald (delta)              1  [0.44,     1]   w = 0.0701
ILR, mapped back       0.33  [0.061,  0.79]   w = 0.0825
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.